

The Cost of Coming Out^{*}

Enzo Brox[†]

Riccardo Di Francesco[‡]

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Abstract

The fear of social stigma leads many individuals worldwide to hesitate in disclosing their sexual orientation. Since concealing identity is costly, it is crucial to understand the extent of anti-LGB sentiments and reactions to coming out. This paper uses an innovative data source from a popular online game together with a natural experiment to overcome existing data and endogeneity issues. We exploit exogenous variation in the identity of a character to identify the effects of coming out on players' revealed preferences for that character across diverse regions globally. Our findings reveal a substantial and persistent negative impact of coming out.

Keywords: LGB economics, social stigma, concealable stigma.

JEL Codes: J15, J71, K38

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[†] University of Bern, Center for Research in Economics of Education and University of St.Gallen, Swiss Institute for Empirical Economic Research. Electronic correspondence: enzo.brox@unibe.ch.

[‡] University of Southern Denmark, Department of Economics, Econometrics and Data Science. Electronic correspondence: rdif@sam.sdu.dk.

1 Introduction

The fear of social stigma significantly shapes human behavior, steering individuals to conceal actions, behaviors, or identity (Goffman, 1956; Bharadwaj et al., 2017). One important part of identity that has increasingly captured public and academic attention over the last two decades is an individual’s sexual orientation (see, e.g., Badgett et al., 2023). Despite significant progress in advancing lesbian, gay, and bisexual (LGB) rights, pervasive anti-LGB sentiments persist in many countries, leading many individuals worldwide to hesitate in openly disclosing their sexual orientation due to lingering fears (Badgett, 2020) and anticipated discrimination (Aksoy et al., 2023).

Existing research indicates that concealing one’s identity comes with a cost (Akerlof & Kranton, 2000). For instance, the stress of hiding one’s sexual orientation is one of the potential *minority stressors* (Meyer, 1995, 2003) that explain the higher prevalence of mental disorders among LGB individuals compared to their heterosexual counterparts (see, e.g., Pachankis et al., 2020). Therefore, understanding the extent of stigma and the challenges faced by individuals upon coming out is crucial to fostering the groundwork for supportive work and living environments (Badgett et al., 2021).

In this paper, we use an innovative data source from an online video game together with a natural experiment to assess the stigma attached to sexual minority status by observing individuals’ responses to sexual minority disclosure. Existing studies examining anti-LGB sentiments, which we review below, usually face at least one of two limitations. First, they rely on a selection-on-observables identification strategy that is likely to suffer from omitted variable bias. Second, they depend on survey data where individuals have the discretion to choose whether to disclose their sexual orientation, thereby introducing additional endogeneity issues (Coffman et al., 2017; Ham et al., 2024). Our use of video game data effectively avoids these selection concerns, and the natural experiment we leverage allows for a credible identification of the responses to revealing sexual minority status.

We use a rich data set from one of the most popular online video games, *League of Legends*. In this game, before a match starts, players are required to choose one playable character. Each character is characterized by game-relevant attributes and a background story that provides details about their history, origin, and relationships with other in-game characters. We leverage an unexpected change in the background story of a playable character, which discloses its

sexual orientation minority status, to examine individuals' responses to sexual minority status disclosure.

Specifically, at the beginning of the 2022 LGBT Pride Month, the game developers announced that one of their playable characters is gay. This event introduces exogenous variation in the character's identity, providing a unique opportunity to examine reactions to sexual minority disclosure. We demonstrate that the announcement was not anticipated, thereby strengthening the credibility of our identification strategy. We then utilize detailed daily data to track players' revealed preferences for the character over a meaningful period. To isolate the effects of the disclosure on players' preferences from potential confounding influences, we employ synthetic control methods to construct a synthetic character closely resembling the pre-announcement preference history of our treated character (see, e.g., Abadie, 2021; Abadie & Vives-i-Bastida, 2022).

Our findings reveal a substantial and persistent negative impact of coming out on players' preferences for the treated character, with a decline of more than 40% of the pre-treatment average preferences for that character. This result consistently holds across various robustness checks. To further assess the external validity of our findings, we conduct a heterogeneity analysis based on players' skill levels. The decline in preferences is equally pronounced among both lower- and higher-skilled players, indicating that the behavioral response is widespread rather than confined to a particular segment of the player base. Additionally, we exploit another unique feature of our setting: the online video game is played globally on different regional servers under very comparable circumstances. We make use of the information on the regional servers to compare how preferences for the treated character evolve across diverse regions across the world. The results consistently demonstrate a negative response across regions.

To strengthen the credibility of preferences towards LGB status as the primary explanation for the estimated effects, it is crucial to ensure that players' decisions to switch from the character are not influenced by other factors. We address and eliminate several alternative channels, thereby enhancing the plausibility of social stigma as the primary explanation for the observed behavior. First, we rule out the possibility that shifts in characters' relative strengths could explain our estimated effect. Second, we show that players' skills have no correlation with the choice to drop the character, thus dismissing the possibility that gameplay factors are the driving force behind the players' observed behavior. Additionally, we demonstrate that players are not leaving the game after the disclosure but are shifting their focus to other characters.

Third, we provide evidence that switching to other characters does not affect the performance of the players involved, highlighting that the decision to abandon the character is not driven by performance considerations. Fourth, we dismiss the possibility that the release of a new character after the disclosure explains the estimated effect.

Finally, we exploit the presence of other playable characters with sexual minority status to show that LGBT Pride Month, which started on the day of the announcement, is unlikely to explain our findings. To do so, we introduce a theoretical framework that formalizes the existence of two “simultaneous treatments”—the disclosure of the character’s sexual orientation and the start of LGBT Pride Month. We outline sufficient assumptions that enable us to separate the impacts of these treatments on players’ preferences for the character.¹ The empirical results support the interpretation that the estimated effects are driven by the character’s disclosure.

While our setting is unconventional, it presents unique advantages for the identification of sentiments towards LGB identity. First, video games offer a controlled research environment, enabling the observation of behaviors that might be challenging to capture through traditional methods (Palacios-Huerta, 2025).² Second, they allow us to leverage objective measures of behavior and identity, circumventing the limitations associated with self-reported identity in surveys (Coffman et al., 2017; Ham et al., 2024). Third, online gaming platforms offer the benefit of anonymity, minimizing social desirability bias and increasing the likelihood of individuals disclosing sensitive information such as their true attitudes toward sexual minority groups.

Despite these advantages, our study’s findings should be interpreted with three important caveats. First, the sample of participants in our setting is unlikely to be representative of the general population. The demographic composition of online gaming communities may differ systematically from broader societal distributions, potentially affecting the external validity of our findings. Second, our analysis focuses on responses to a male character revealing his sexual minority status. Given prior evidence of differential labor market discrimination against gay men and lesbian women (Badgett et al., 2023), it is plausible that reactions would vary if a female character disclosed the same identity. Third, our study examines preferences for a fictitious character rather than an actual human being. As a result, factors such as the character’s perceived prestige or popularity may influence individuals’ responses, making our

¹ See, e.g., Roller and Steinberg (2023) for a discussion on “simultaneous treatments” and methodologies for disentangling their effects under a difference-in-differences identification strategy.

² A similar argument underlies several studies that use data from the professional sports industry to examine, for example, discrimination (Price & Wolfers, 2010; Pope et al., 2018), incentive effects (Pope & Schweitzer, 2011), and other behavioral patterns (Brox & Goller, 2025).

setting most analogous to the context of supporting a well-known public figure, such as a politician or professional athlete.

Our paper contributes to three distinct strands of the literature. First, it relates to a growing literature studying the economics of LGBTQ+ individuals (see, e.g., Badgett et al., 2021, 2023). The current body of research primarily focuses on measuring discrimination against LGB individuals by comparing their labor market outcomes with those of non-minority individuals with similar observable characteristics, mostly documenting wage penalties for gay and bisexual men and wage premiums for lesbian women (see, e.g., Badgett, 1995; Carpenter & Eppink, 2017; Martell, 2021; Carpenter et al., 2023).³ However, despite their valuable contributions, these studies suffer from the endogeneity and selection issues discussed above that hinder the ability to draw causal inferences from their findings.⁴ To tilt towards a more causal interpretation, other studies use correspondence designs to probe into hiring discrimination against LGB individuals, consistently revealing that LGB job candidates are less likely to be invited for interviews or offered job opportunities (Weichselbaumer, 2003; Drydakis, 2009; Tilcsik, 2011; Ahmed et al., 2013; Drydakis, 2014). Nevertheless, a significant challenge lies in communicating that an individual belongs to a sexual orientation minority group, since such information is not typically included in job applications. This raises the question of whether the observed results are affected by the choice and nature of the signal used (Bertrand & Duflo, 2017).

We make several contributions to this literature. First, our use of data from an online video game allows us to overcome the endogeneity issues of previous studies and to identify preferences toward LGB status. Second, to the best of our knowledge, our study is the first to investigate the immediate reactions to coming out. Due to the substantial anecdotal and scientific evidence that individuals often hide their sexual orientation status (see, e.g., Badgett, 2020) and the prevalent evidence of the associated costs (Meyer, 2003; Pachankis et al., 2020), several studies have explicitly focused on investigating the determinants and incentives of coming-out decisions.⁵

³ The only study finding a wage premium for gay men is that of Carpenter and Eppink (2017). Tampellini (2024) does not find a wage penalty but finds a lower probability of being full-time employed and a higher probability of being a victim of work-related violence for gay men. There is also evidence that transgender workers face earning and employment penalties (Geijtenbeek & Plug, 2018; Carpenter et al., 2022).

⁴ A related literature studies whether changes in laws and norms affect labor market outcomes and attitudes towards LGB individuals (see, e.g., Burn, 2018; Ofosu et al., 2019; Sansone, 2019; Burn, 2020; Delhommer, 2020; Deal, 2022, 2023). Broockman and Kalla (2016) show that a randomized intervention that encourages actively taking the perspective of others can reduce transphobia.

⁵ Seror and Ticku (2021) investigate the impact of same-sex marriage legalization on coming-out decisions using a revealed preference mechanism inferred from data on Catholic priests' vow of celibacy, finding reduced demand for priestly studies after the adoption of same-sex legalization. Gromadzki and Siemaszko (2022) explore spillover effects of coming-out decisions using Twitter data, discovering positive externalities, as exposure to peers coming out is associated with a higher probability of individuals coming out themselves. Aksoy et al.

Our study complements these studies by focusing on responses to revealing sexual minority status, potentially providing a rationale for the observed underrepresentation of individuals with sexual minority status in many areas.

Second, our findings relate to the broader literature on discrimination. Early work by Becker (1957) paved the way for an extensive literature investigating discrimination instances, mostly based on gender and ethnicity, across diverse economic domains (see, e.g., Kuhn & Shen, 2013; Arnold et al., 2022). A substantial portion of the empirical evidence stems from field experiments, in which researchers use audit and correspondence studies to isolate the causal impact of identity on behavior (see, e.g., Ayres & Siegelman, 1995; Oreopoulos, 2011).⁶ Our study contributes by exploring sentiments against sexual orientation minorities. The concealable nature of sexual orientation allows individuals to anticipate discrimination and strategically choose to hide their identity (Aksoy et al., 2023). This complicates the use of traditional methods to investigate discrimination against individuals based on their sexual orientation and necessitates innovative strategies for understanding sentiments against groups with stigmatized identities, particularly in environments prone to discrimination.

Third, our study contributes to the literature on video games and social interaction platforms (Ederer et al., 2024). The recognition of video game data’s potential for research is increasing among economists, who are already capitalizing on the abundance and quality of the available data.⁷ However, despite its industry size and the significant fraction of time spent on playing video games, the scientific literature on these social platforms is only slowly increasing.⁸ Parshakov et al. (2023) investigate consumer behavior in the video game industry. They examine the impact of marking products with a gay label on consumers’ demand, finding a significant, albeit short-lived, decrease in demand following the introduction of the gay label. Gandhi et al. (2024) study entertainment preferences in the context of video games and test economic theories of belief-based utility using data from League of Legends. Our study complements this

(2023) conduct a lab experiment demonstrating that individuals strategically hide their sexual orientation in anticipation of discrimination in prosocial behavior, a result consistent with that of Kudashvili and Lergetporer (2022).

⁶ For recent surveys, see Bertrand and Duflo (2017) and Neumark (2018). Onuchic (2022) provides a detailed review of traditional statistical and taste-based discrimination models, along with a discussion of recent theories that expand on these models.

⁷ For example, Parshakov et al. (2018) explore the impact of diversity on team performance using data from an online video game, and Dell’Acqua et al. (2023) investigate the effects of artificial intelligence on collaborative production dynamics in controlled laboratory settings using a team-based video game. Correll et al. (2002) utilize a self-designed video game to study racial discrimination.

⁸ Aguiar et al. (2012) highlight that, according to the American Time Use Survey, almost 10 % of leisure time is spent on playing games and computer use. Global revenues of the video game industry were around \$180 billion in 2020.

line of research by expanding the understanding of individuals’ behavior within these virtual environments.

The rest of the paper unfolds as follows. Section 2 describes the key elements of League of Legends that are relevant to our study and outlines the natural experiment we leverage to identify the effects of coming out. Section 3 introduces the data and explains the methodology we use to isolate the effects of coming out. Section 4 presents our main results. Section 5 examines the underlying mechanisms driving the estimated effects. Section 6 concludes.

2 Context

In this section, we explore the contextual framework that enables us to measure reactions to the disclosure of sexual orientation. Specifically, we turn our attention to the online video game *League of Legends* as our data source and the natural experiment we leverage to credibly identify the causal effects of coming out.

The next subsection describes the key elements of League of Legends that are relevant to our study. Our main analysis does not rely on in-game information, but instead focuses on the pre-match phase. Therefore, we do not provide an exhaustive account of how matches unfold, but rather emphasize the details that inform our research. Then, we discuss the coming-out event we exploit and its implications for identification purposes.

2.1 League of Legends

League of Legends is a globally prominent multiplayer online game developed and published by Riot Games. Originally launched on October 27th, 2009, the game attracted a vast player base, averaging 180 million monthly active players as of 2022 and peaking at 14 million players in a single day. It has also achieved substantial financial success, generating \$1.8 billion in revenue in that same year.⁹

League of Legends demands significant time and cognitive engagement from its user base. In our sample alone—which follows a subset of players over a seven-month period—users collectively spent 647,966 hours in matches. Valuing this gameplay time at the 2022 U.S. federal minimum wage of \$7.25 per hour implies an aggregate opportunity cost of \$4.7 million. This figure represents a conservative lower bound of time spent on the game, as it excludes substantial

⁹ See, e.g., <https://prioridata.com/data/league-of-legends>.

time spent on pre- and post-match activities.

In League of Legends, players are divided into two teams of five players each to compete in matches with the aim of destroying the opposing team’s base. Players in each team sort themselves into one of five roles. These roles represent crucial strategic positions, each requiring specific playstyles and contributing differently to the team’s final objective.

Before a match begins, players must select a playable character to control during the match. In our analysis, we measure players’ revealed preferences for a specific character by quantifying how frequently they select that character for their matches. Our objective is to investigate whether these preferences undergo any shifts following the disclosure of the character’s sexual orientation. Thus, we devote the rest of this section to exploring the design of characters in League of Legends and the process through which players select their characters for matches.

Each character has a unique set of skills and abilities and is specifically designed to excel in one or two of the distinct roles that players can take on within the team.¹⁰ Additionally, characters are crafted with a rich background that adds a narrative dimension to the game but does not have any impact on the game’s mechanics. This is achieved through the creation of detailed biographies and short stories that provide players with a deeper understanding of the character’s history and motivations, thus offering players the opportunity to connect with their chosen characters on a more personal level.¹¹

The character selection process occurs in a virtual lobby where players can communicate with their teammates through a chat function. In a random order that alternates between teams, players take turns selecting their characters for the match. Once a player chooses a character, their selection becomes visible to all players participating in the match, including the opposing team. Each character can be chosen by only one player, making it unavailable for selection by others. Once all players have selected their characters, the match begins.

When making their character selection, players consider various factors. First, they consider the role they are assigned to fulfill in the game. Each role has its own set of responsibilities and playstyle requirements, and players aim to choose a character that aligns with their designated role. Second, players take into account their personal mastery of specific characters, opting for those they are most skilled and comfortable with. Third, players may also consider their personal preferences, such as the playstyle and background story of the character, adding a

¹⁰ These roles are labeled *top*, *jungle*, *mid*, *bottom*, and *support*.

¹¹ The list of all characters along with details about their abilities and histories is available at <https://www.leagueoflegends.com/en-gb/champions/>.

subjective element to the selection process.

2.2 Coming-Out Event

Every year in June, *LGBT Pride Month*, a dedicated time to honor and celebrate the LGBT community, takes place. Since 2018, Riot Games has actively participated in this month-long celebration by integrating new content into League of Legends during the month of June. This includes the introduction of in-game cosmetics, such as character skins, as well as emotes that allow players to express themselves in the game. It is important to note that while these additions enhance the visual and expressive elements of the game, they do not alter the game’s mechanics or the characteristics and abilities of the League of Legends characters.¹²

Riot has also supported the representation of the LGBT community in the game by unveiling the sexual minority status of specific characters. In our paper, we primarily focus on the character *Graves*, chosen due to the existence of a well-defined announcement regarding his sexual orientation.¹³

At the beginning of the 2022 LGBT Pride Month, Riot Games released a short story featuring Graves and *Twisted Fate*. This story officially unveils Graves’ sexual orientation, establishing him as a gay character.¹⁴ The following quotes provide two pivotal passages of the narrative:¹⁵

I do not have terrible taste in men. I have good taste in terrible men. (Graves)

[...] asked Fate with a tinge of poorly concealed jealousy, despite Graves having been gay for the better part of four decades. (Storyteller)

This *coming-out event* closely approximates an ideal experiment where individuals randomly disclose their sexual minority status, thus providing a unique setting to identify the effects of coming out on players’ preferences for Graves.¹⁶

¹² We check this in Section 5.1, where we demonstrate that characters’ strength was unaffected by LGBT Pride Month.

¹³ For instance, in 2018, the character *Neeko* was confirmed as lesbian, marking her as the first openly LGBT character in the game. Furthermore, in 2021, the characters *Diana* and *Leona* were revealed to be lesbians, while the character *Nami* was declared to be lesbian and polyamorous.

¹⁴ The story unveiling Graves’ sexual orientation also subtly hints at Twisted Fate’s pansexuality, although this is not explicitly stated. We investigate whether this implied revelation has captured the players’ attention in Appendix C. We highlight the relatively low attention directed towards Twisted Fate from players, who were primarily focused on Graves and the explicit establishment of his sexual orientation. As a result, we concentrate our analysis on Graves and his disclosure for a more credible identification of the effects of coming out.

¹⁵ The whole story is available at https://universe.leagueoflegends.com/en_SG/story/the-boys-and-bombolini/.

¹⁶ It is crucial to distinguish between the *coming-out event* and the disclosure of Graves’ sexual orientation. The coming-out event encompasses both Graves’ disclosure and the start of LGBT Pride Month. While this is

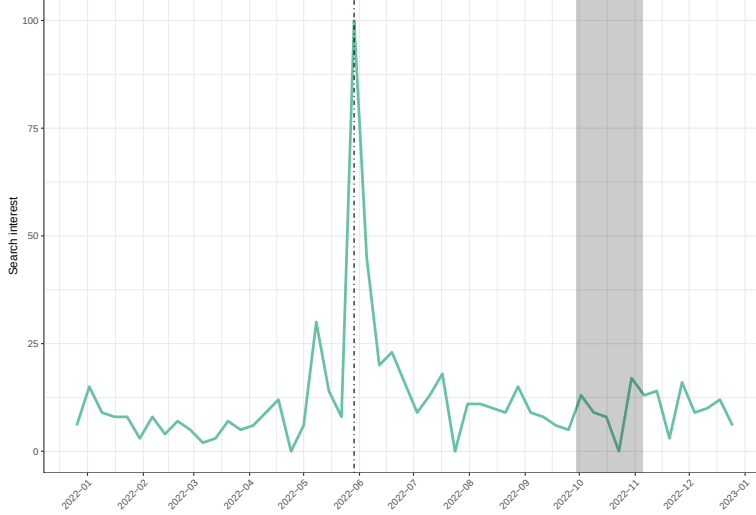


Figure 1: Google search interest over time for the query “*Graves gay.*” The dashed vertical line denotes the week of disclosure, and the shaded area highlights the League of Legends World Championship.

To ensure the credibility of our identification, it is crucial that the disclosure was not anticipated by players. Figure 1 displays the Google search interest for the query “*Graves gay.*” We observe minimal interest in this search term throughout the year 2022, with a remarkable spike occurring during the week of the coming-out event. This pattern supports our assumption of no anticipation and strengthens the credibility of our identification strategy. What is particularly interesting is that this surge in interest surpasses the level observed during the 2022 League of Legends World Championship (held from September 29th to November 5th), despite Graves being among the top-eight most played characters during the tournament. This finding emphasizes the substantial impact and attention that the coming-out event received from players.

3 Data and Methodology

In this section, we introduce the data and explain the methodology employed in our analysis. The next subsection outlines the construction of our data sets and details how we gauge players’ revealed preferences for characters. We then provide a formal review of the synthetic control estimators employed to isolate the effects of the coming-out event on players’ preferences for Graves.

not a concern for identification, it requires careful interpretation of the findings. To maintain clarity, we generally refer to the effects of the coming-out event in our analysis. Further discussion on this topic is deferred to Section 5.5 and Appendix D.

3.1 Data

We obtain our data by accessing the Riot Games API, which provides us with valuable information about League of Legends matches.

The game operates on multiple servers located worldwide. Within these servers, we target the top tier of the League of Legends ranked system. By targeting this specific group of players, we aim to minimize the noise that may arise from players who are not fully engaged in the game, thus reducing the risk of attenuation bias.¹⁷ Furthermore, this focus increases the chances that players in our data set are aware of the coming-out event, thereby strengthening the credibility of our identification strategy.¹⁸

Specifically, we identify the players within the top tier of each server as of July 2022. For each of these players, we gather raw data on the full history of matches they engaged in during the period January-July 2022. This approach inherently includes, to a large extent, players who were not in the top tier as of July 2022 but have been matched with our focal players during the time span we consider.

To clean the raw data, we apply a multi-step procedure. First, we aggregate server-specific regions into four primary macro-regions—Latin America, North America, Europe, and Korea—and exclude servers where match coverage is sparse, namely Japan, Oceania, Russia, and Turkey.¹⁹ Second, we eliminate duplicate observations, that arise when the same match appears multiple times due to its inclusion in the histories of multiple players. We also drop corrupted matches, defined as those with fewer than ten unique participants, a duration exceeding two hours, or missing data. Finally, we assign each character “main” and “auxiliary” roles based on historical usage patterns and exclude matches in which characters were used in atypical roles.²⁰

After these steps, our clean sample comprises 136,399 unique players participating in 146,451

¹⁷ Players can choose between *draft* or *ranked* matches. Both game modes share the same mechanics and objectives. However, while draft matches are more casual and do not have consequences for players’ rankings or ratings, in ranked matches players earn or lose points based on the outcome of the match to determine their position within the ranked system. By focusing on ranked matches we aim to further reduce the risk of attenuation bias.

¹⁸ We acknowledge that some of these players may have followers or engage in streaming activities, which could reduce the level of anonymity within our sample. Essentially, our focus trades off a degree of anonymity for reduced attenuation bias and strengthened credibility of identification.

¹⁹ The Latin America macro-region includes servers from North Latin America, South Latin America, and Brazil. Europe combines the North/East Europe and West/South Europe servers. Korea and North America are standalone servers. Further information on server structure and geographic coverage is available at <https://wiki.leagueoflegends.com/en-us/Servers>.

²⁰ Characters in League of Legends are explicitly designed to fulfill a primary and a secondary role. We use observed historical play patterns to empirically assign each character their main and auxiliary roles. Excluding matches where characters are used outside these roles ensures that our measure of character usage reflects meaningful player choices, rather than experimental or off-meta behavior that could introduce noise into the analysis.

matches.

From the clean data, we construct a character-level balanced panel that tracks the daily usage of each character across all major regions. Our final sample includes 161 characters tracked over 194 days. To gauge players’ revealed preferences for characters, we construct a metric called *pick rate*, which measures the frequency with which players choose a specific character in their games each day in each region. Our primary objective is to investigate whether the disclosure of Graves’ sexual orientation influences the pick rate of this character. We further compute daily *win rates*, measuring the fraction of matches in which each character is on the winning team relative to the total number of matches that character participates in on a given day. This serves as a proxy for character’s strength, allowing us to test whether shifts in character popularity are driven by changes in effectiveness rather than preferences alone.

We further construct an additional player-level daily panel that captures individual behavior over time. To ensure meaningful variation, we restrict the sample to players who participated in at least 10 matches both before and after the coming-out event, resulting in 7,421 unique players. For each player, we compute the number of matches played and hours spent in-game each day, along with their daily win rates.²¹ We also record the frequency with which players select specific characters and roles. This data set enables us to examine whether individual behavior adjusts in response to the coming-out event.

Table 1 presents descriptive statistics on Graves usage and player behavior before and after the coming-out event. Graves’ pick rate declines from 18.56% to 12.32%—a drop of more than 30%—suggesting a substantial shift in player preference for Graves.²² This decline is not accompanied by a deterioration in Graves’ win rate, which remains essentially flat. We further see no evidence that players disengaged from the game following Graves’ disclosure. On the contrary, both the average number of daily matches and hours spent in-game slightly increase post-treatment, ruling out attrition or general disengagement.²³ Finally, players’ role selection patterns remain largely stable across the pre- and post-treatment periods. The lack of major shifts suggests that players who stopped using Graves likely substituted toward other characters within the same role (*jungle*). We investigate the possibility of intra-role spillovers in greater

²¹ Hours spent in-game capture only the duration of matches. This substantially underestimates total time spent on the game, as it excludes time devoted to matchmaking, character selection, and other pre- and post-match activities.

²² Table A.I in Appendix A shows that, conditional on Graves’s primary role, it ranks as the second most frequently selected character before the disclosure. However, after the treatment, its ranking drops to fifth place.

²³ Figure A.I in Appendix A illustrates the distribution of players’ total and average daily activity, measured in both number of matches played and hours spent in-game.

Table 1: Pre- and post-treatment summary statistics.

	Pre-treatment	Post-treatment	Mean difference
Graves metrics			
Graves pick rate (%)	18.56 (3.81)	12.32 (3.13)	-6.24
Graves win rate (%)	50.19 (4.82)	49.04 (4.21)	-1.15
Player activity			
Win rate (%)	51.51 (42.08)	50.47 (40.47)	-1.04
N. daily matches	2.03 (1.62)	2.27 (1.75)	+0.24
Time spent in-game (hours)	0.89 (0.72)	1.01 (0.78)	+0.11
Role selection patterns			
Top (%)	18.51 (36.63)	18.86 (36.67)	+0.35
Jungle (%)	20.93 (38.58)	20.83 (38.09)	-0.11
Mid (%)	19.68 (36.89)	19.18 (36.10)	-0.50
Bottom (%)	20.13 (38.08)	21.21 (38.35)	+1.09
Support (%)	20.74 (38.84)	19.91 (37.77)	-0.83

Notes. Sample averages are reported for each variable, followed by standard deviations in parentheses. Graves' pick and win rates are computed from the character-level data; remaining statistics are computed from the player-level data. Time spent in-game only captures time within matches, excluding character selection and matchmaking time. The final column reports differences in means between the pre- and post-treatment periods.

detail in our robustness checks and find that our main results are unaffected.

3.2 Methodology

A simple comparison of Graves' pick rates before and after the disclosure may not accurately reflect the impact of the coming-out event on players' preferences for that character, as other factors could have changed during that period. To address this issue, we construct a synthetic control unit (see, e.g., Abadie & Gardeazabal, 2003; Abadie et al., 2010, 2015; Abadie, 2021; Abadie & Vives-i-Bastida, 2022) by weighting other characters to approximate the pick rates of Graves before the disclosure. This method allows us to isolate the effects of the coming-out event on players' revealed preferences for Graves and gain insight into how these preferences would have behaved in the absence of the disclosure.

Formally, our data set comprises $n = 161$ characters ($i = 1, \dots, n$) observed over $T = 194$ days ($t = 1, \dots, T$). T^{pre} is the length of the period (150 days) before the coming-out event, which occurs at time $T^{pre} + 1$ (i.e., June 1st, 2022). For each unit i and time t , we denote the observed pick rate as $Y_{i,t}$. We represent the coming out as a binary variable $C_i \in \{0, 1\}$ equal to one if

character i discloses his sexual orientation at time $T^{pre} + 1$. We then posit the existence of two potential pick rates $Y_{i,t}^c$, where one denotes the pick rate in the absence of disclosure ($Y_{i,t}^0$) and the other denotes the pick rate in the presence of disclosure ($Y_{i,t}^1$).²⁴

Without loss of generality, we let the first unit $i = 1$ be Graves. This implies that $C_1 = 1$ and $C_i = 0$ for all $i \neq 1$. Then, for each period $t > T^{pre}$, we define the effect of the coming-out event on players' preferences for Graves as the difference in Graves's potential pick rates at time t :

$$\tau_t := Y_{1,t}^1 - Y_{1,t}^0. \quad (1)$$

Note that we allow the effects to change over time.

Since Graves' sexual orientation has been disclosed after period T^{pre} , under a standard SUTVA assumption (see, e.g., Imbens & Rubin, 2015) we observe $Y_{1,t} = Y_{1,t}^1$ for all $t > T^{pre}$. Thus, as shown in equation (1), the challenge in estimating our causal effects of interest is to estimate $Y_{1,t}^0$ for $t > T^{pre}$, i.e., how Graves' pick rates would have evolved in the absence of the disclosure. To this end, we can construct a synthetic control unit that approximates the pick rates of Graves before the coming out. The idea is that if the synthetic control and Graves behave similarly before the disclosure, then the synthetic control can serve as a valid counterfactual.

The synthetic control unit is characterized by a set of weights, denoted as $\omega := (\omega_2, \dots, \omega_n)$, chosen to align the pre-treatment pick rates of the synthetic unit with those of Graves. This is achieved by solving the following optimization problem (Arkhangelsky et al., 2021):

$$\begin{aligned} \hat{\omega} &= \arg \min_{\omega \in \Omega} \ell(\omega), \\ \ell(\omega) &= \sum_{t=1}^{T^{pre}} \left(\sum_{i=2}^n \omega_i Y_{i,t} - Y_{1,t} \right)^2 + \zeta^2 T^{pre} \|\omega\|_2^2, \quad \Omega = \left\{ \omega \in \mathbb{R}_+^{n-1} : \sum_{i=2}^n \omega_i = 1 \right\}, \end{aligned} \quad (2)$$

where the weights are restricted to be non-negative and to sum up to one and a ridge penalty is employed to ensure the uniqueness of the weights. In our main specification, we set the regularization parameter to zero, thus employing a standard synthetic control estimator. As a robustness check, we follow Arkhangelsky et al. (2021) and set $\zeta = (T - T^{pre})^{1/4} \hat{\sigma}$, with $\hat{\sigma}$ denoting the standard deviation of first differences of $Y_{i,t}$ for control units over the pre-treatment period.

²⁴ These potential outcomes are based on Rubin's model for causal inference (Rubin, 1974).

We estimate the counterfactual outcome of Graves as a weighted average of the outcome of the control units:

$$\hat{Y}_{1,t}^0 = \sum_{i=2}^n \hat{\omega}_i Y_{i,t}. \quad (3)$$

Finally, to estimate the causal effects of interest, we compute the differences between Graves’ observed pick rates and the synthetic counterfactual for all $t > T^{pre}$:

$$\hat{\tau}_t = Y_{1,t}^1 - \hat{Y}_{1,t}^0. \quad (4)$$

We summarize the estimated effects by reporting the average treatment effect on players’ preferences for Graves, with the averaging carried out over the post-treatment periods:

$$\hat{\tau} = \frac{1}{T - T^{pre}} \sum_{t=T^{pre}+1}^T \hat{\tau}_t. \quad (5)$$

We employ the “placebo approach” of Arkhangelsky et al. (2021) to estimate the variance of $\hat{\tau}$. We then use the estimated variance to construct asymptotically valid conventional confidence intervals.²⁵

4 Results

In this section, we present our main results. First, we present our main findings. We then investigate heterogeneity across players with different levels of in-game performance. Finally, we explore the possibility of regional variations in attitudes toward the LGB community by replicating our analysis across different macro-regions. Appendix B provides a series of robustness checks that support the reliability of our estimates.

4.1 Main Results

We apply the synthetic control estimator of Section 3.2 to estimate the effects of the coming-out event on players’ revealed preferences for Graves. To mitigate the potential for spillover effects, we exclude Twisted Fate and other four characters (*Diana*, *Leona*, *Nami*, and *Neeko*) that were already members of the LGB community prior to the coming-out event from the donor pool.²⁶

²⁵ The validity of this placebo approach hinges on a homoskedasticity assumption which requires that treated and control units have the same noise distribution. In general, with only one treated unit, nonparametric variance estimation for treatment effect estimators is typically impossible without a homoskedasticity assumption (Arkhangelsky et al., 2021).

²⁶ Nevertheless, even if included in the donor pool, the estimator assigns these characters zero weight.

Figure 2 displays Graves’ actual and synthetic pick rate series.²⁷ Overall, our analysis suggests a substantial negative impact of the coming-out event on players’ preferences for Graves. Before the disclosure, the synthetic control estimator closely approximates the trajectory of Graves’ pick rates, providing support for the estimator’s ability to predict the counterfactual series. However, starting from June 1st, 2022, the two series diverge substantially, with Graves’ pick rates consistently dropping below those of the synthetic control. This gap persists over time, extending even beyond the conclusion of LGBT Pride Month. The estimated average effect, reported in the first column of Table B.I in Appendix B, is -7.60 percentage points and is statistically different from zero at the 5% significance level.²⁸ This implies a decline of 40.98% of the pre-treatment average preferences for Graves.

We conduct a series of robustness checks, detailed in Appendix B, all of which reinforce our main finding. First, our results are robust to alternative estimation strategies and donor pool compositions: we obtain consistent estimates using a regularized synthetic control estimator and when restricting the donor pool to characters unlikely to act as substitutes for Graves. Second, we assess predictive accuracy by artificially backdating the treatment (see, e.g., Abadie & Vives-

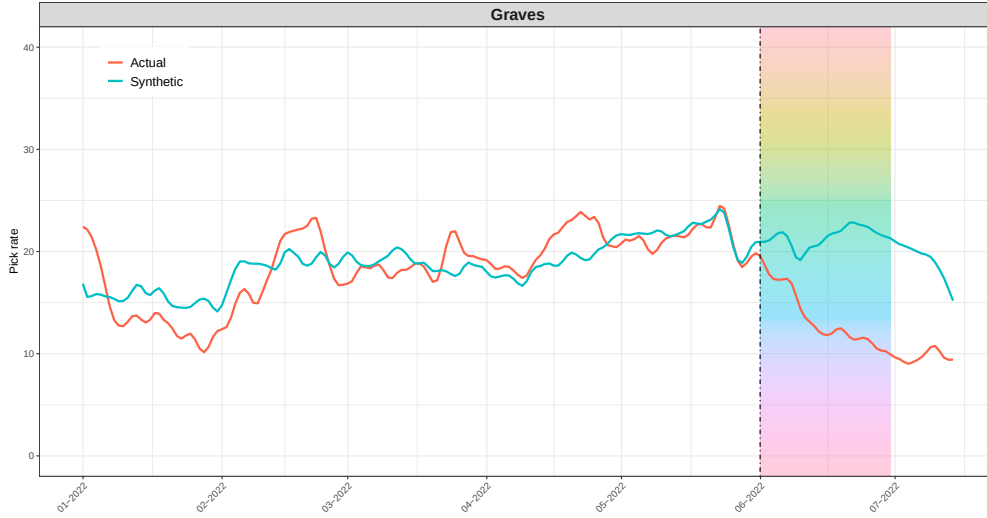


Figure 2: Graves’ daily pick rates and synthetic control estimation results. The actual series is smoothed by a Nadaraya-Watson regression before applying the synthetic control estimator. The dashed vertical line denotes the day of disclosure, and the rainbow area highlights LGBT Pride Month.

²⁷ Figure A.II in Appendix A displays the identities and the contributions of the characters in the donor pool with non-zero estimated weights.

²⁸ To provide further evidence that our main estimate is not driven by chance, we reassign the treatment to all control characters in our data set. This allows us to generate a distribution of “placebo” effects, which serves as a basis for comparing the impact on Graves (see, e.g., Abadie et al., 2010, 2015). We report the ratios of post-to pre-treatment root mean squared error (RMSE) in Figure A.III in Appendix A, where Graves ranks third in terms of RMSE ratio. This implies that the impact on Graves is unusually large compared to the distribution of placebo effects, reinforcing the interpretation that our analysis provides significant evidence of a negative effect of the coming-out event.

i-Bastida, 2022); discrepancies between the actual and synthetic series emerge only on the true disclosure date and closely match those in the main analysis, demonstrating the estimator’s ability to forecast the counterfactual series and supporting a no-anticipation assumption. Third, a leave-one-out analysis (see, e.g., Abadie, 2021) confirms that no single donor character drives the results.

4.2 Player Heterogeneity

To examine whether responses to the coming-out event differ by player ability, we conduct a heterogeneity analysis based on a proxy for player skill derived from pre-treatment in-game performance. Specifically, we compute each player’s win rate prior to the treatment date, restricting the sample to the same 7,421 users selected for our player-level data set to ensure the reliability of this metric. We then classify these players into above- and below-median skill groups based on this metric. Using this stratification, we replicate our main synthetic control analysis separately for each group. The estimation proceeds identically to our baseline specification of Section 4.1.

This stratification allows us to investigate whether the decline in Graves’ pick rate is concentrated among particular subpopulations—such as lower-skilled or more casual players—or whether it reflects a broader behavioral response. Consistent estimates across skill groups would strengthen the external validity of our findings, mitigating concerns that the observed response is driven by a narrow or non-representative segment of the player base.

Figure A.IV in Appendix A plots the distribution of overall win rates and total matches for players in each skill group. Although the win rate distributions overlap substantially, they differ in their means, with players classified as higher-skilled exhibiting slightly higher win rates on average. In contrast, the distribution of total matches is nearly identical across groups, indicating that conditioning on pre-treatment win rate does not mechanically segment players by engagement level. This is important for interpretation: it implies that our stratification holds constant (approximately) overall game activity while isolating variation in a meaningful proxy for player quality.

Figure 3 presents the results. We observe a sustained decline in Graves’ pick rate immediately following the coming-out event in both the below- and above-median skill groups. The timing and persistence of the divergence from the synthetic counterfactual are remarkably similar across the two groups and closely mirror the patterns documented in our pooled speci-

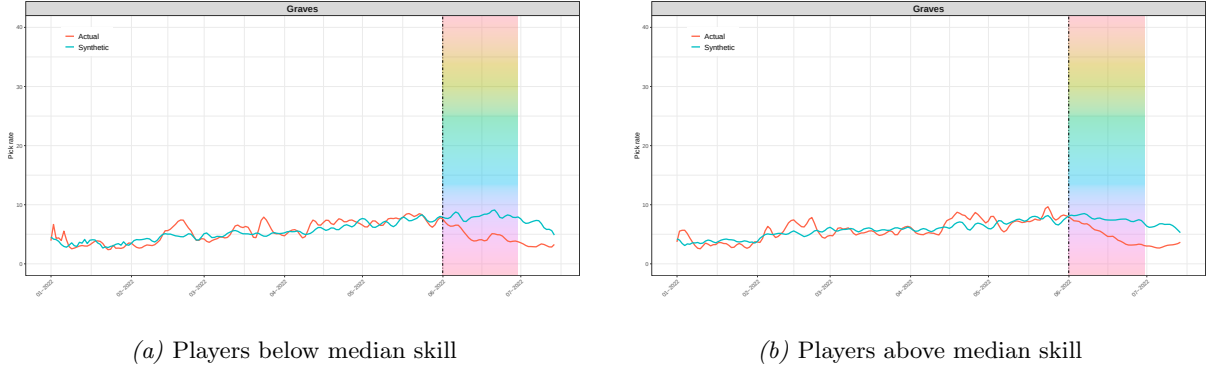


Figure 3: Graves’ daily pick rates and synthetic control estimation results by skill group. The actual series is smoothed by a Nadaraya-Watson regression before applying the synthetic control estimator. The dashed vertical line denotes the day of disclosure, and the rainbow area highlights LGBT Pride Month.

fication.²⁹ These findings indicate that the estimated effect is not concentrated among players of a particular ability level but instead reflects a widespread behavioral response, thereby enhancing the external validity of our findings and mitigating concerns that the observed decline in Graves’ popularity is driven by a narrow or non-representative subset of users.³⁰

4.3 Regional Heterogeneity

Previous research has demonstrated that attitudes toward LGB people can substantially vary between countries, causing also a large variation in the number of individuals openly identifying with the LGB community (see, e.g., Badgett, 2020; Badgett et al., 2021).³¹ To explore regional differences in players’ attitudes towards this community, we exploit the geographic information associated with each match. Specifically, we divide the sample according to the macro-regions defined in Section 3.1 and apply the synthetic control estimator described in Section 3.2 separately within each region.

Figure 4 displays the results.³² The synthetic control estimator closely approximates the trajectory of Graves’ pick rates for matches in Europe and Latin America before the disclosure, exhibiting pre-treatment root mean squared errors comparable to those of the pooled specifications. However, discrepancies arise in Korean and North American matches, where the

²⁹ Appendix B shows that these findings are robust to backdating and leave-one-out exercises.

³⁰ While qualitatively similar, the pre-treatment pick rates of Graves are lower in both skill groups relative to the main specification. This difference arises from the additional sample restriction imposed here, whereby we include only players who participated in at least 10 matches both before and after the coming-out event to ensure a meaningful proxy for skill. The main specification, by contrast, includes the full player base, potentially encompassing more casual users with higher Graves usage. As such, differences in baseline levels should be interpreted as compositional.

³¹ An OECD report shows that even within a set of relatively comparable OECD countries, the size of the LGB communities differs by a factor of four (OECD, 2019).

³² Figure A.V in Appendix A displays the identities and the contributions of the characters in the donor pool with non-zero estimated weights.

pre-treatment root mean squared error is two to three times higher than that achieved with European and Latin American matches. In Europe, Korea, and Latin America, we estimate negative and persistent effects of the coming-out event on players' preferences for Graves. Point estimates for the average treatment effect in these regions, reported in Table A.II in Appendix A, are consistently negative and are statistically significant at the 5% level across most of the considered specifications. In North America, where the pre-treatment root mean squared is relatively large, we are unable to determine the sign of the impact, as the confidence intervals for the estimated average effect consistently encompass zero.

The estimated average effect varies substantially across regions, with the largest effect observed in Europe (-62.23% relative to the pre-treatment average preferences) and the smallest effect observed in Korea (-31.82%) in our main specification, indicating heterogeneous responses to the coming-out event. However, it is important to consider that the different magnitudes of the estimated effects do not necessarily reflect differential attitudes towards the LBG community, as various other factors may differ across servers. One such factor could be the differential levels of competitiveness on different servers, which may affect the character selection process by introducing different levels of subjectivity. We therefore carefully interpret our results in this section as evidence for anti-LGB sentiments in various regions, rather than interpreting the differences in the effect sizes.



Figure 4: Graves' daily pick rates and synthetic control estimation results by region. The actual series are smoothed by a Nadaraya-Watson regression before applying the synthetic control estimator. The dashed vertical line denotes the day of disclosure, and the rainbow area highlights LGBT Pride Month.

5 Mechanisms

In Section 4, we established evidence of a substantial negative impact of the coming-out event on players’ revealed preferences for Graves. However, the players’ decision to switch from this character might be influenced by factors beyond preferences for LGB status. The objective of this section is to eliminate these alternative channels, thereby enhancing the plausibility of social stigma as the primary explanation for the observed behavior.

First, we examine the idea that shifts in character relative strengths could explain our estimated effect. We rule out this possibility in Section 5.1 by demonstrating that Graves’ strength remained unaffected by the coming-out event. Second, we explore the potential influence of players’ skills on their decision to abandon Graves. In Section 5.2, we show that players’ skills have no correlation with the choice to drop the character, thus dismissing the possibility that gameplay factors are the driving force behind the players’ observed behavior. Third, we investigate whether players transitioning away from Graves experience any performance-related consequences. This is the topic of Section 5.3, where we present evidence that switching to other characters does not affect the performance of the players involved. This emphasizes our ability to measure players’ true social attitudes and stigma, avoiding any potential biases stemming from strategic performance considerations. Fourth, in Section 5.4, we dismiss the possibility that the release of a new potentially substitute character drives the players’ decisions to switch away from Graves.

Finally, we acknowledge that questions may arise about whether the findings of Section 4 are solely a consequence of Graves’ disclosure or if they are influenced by the broader context of LGBT Pride Month. We exploit the presence of other playable characters with sexual minority status to show that LGBT Pride Month is unlikely to explain our findings. Specifically, Appendix D introduces a theoretical framework that formalizes the existence of two “simultaneous treatments” and outlines sufficient assumptions that enable us to separate the impacts of coming out and LGBT Pride Month on players’ preferences for Graves. The results, detailed in Section 5.5, support the interpretation that the estimated effects are driven by Graves’ disclosure.

5.1 Graves’ Strength

Crucial to the plausibility of social stigma attached to playing an LGB character as the primary explanation for the players’ observed behavior is the fact that Graves’ strength remained

unaffected by the coming-out event, as any change in character relative strengths could explain why players’ preferences shift away from Graves.

To address this concern, we employ the synthetic control estimator described in Section 3.2 to examine the potential impact of the coming-out event on Graves’ strength. We measure characters’ strength using daily win rates.

Figure 5 displays the results. Overall, our analysis reveals that the coming-out event had no impact on Graves’ strength. Despite the actual series exhibiting daily fluctuations around the 50% mark, the synthetic control estimator effectively captures its pre-treatment trend, showcasing its ability to predict the counterfactual trend. After the treatment date, the synthetic control estimator continues to align with Graves’ win rate trend, confirming that the character’s strength was unaffected by the disclosure. The average effect is estimated to be -1.046 percentage points (standard error: 5.298), and the conventional 95% confidence interval encompasses zero, indicating a failure to reject the null hypothesis of no effect. These findings demonstrate that Graves’ strength remained unchanged during the coming-out event, dismissing the possibility of a shift in his strength as an explanation for the results of Section 4.

Moreover, we note that players have real-time access to detailed information regarding characters’ strengths, weaknesses, and overall performance, as numerous websites continuously provide updated data on characters’ in-game statistics.³³ Therefore, players were well-informed

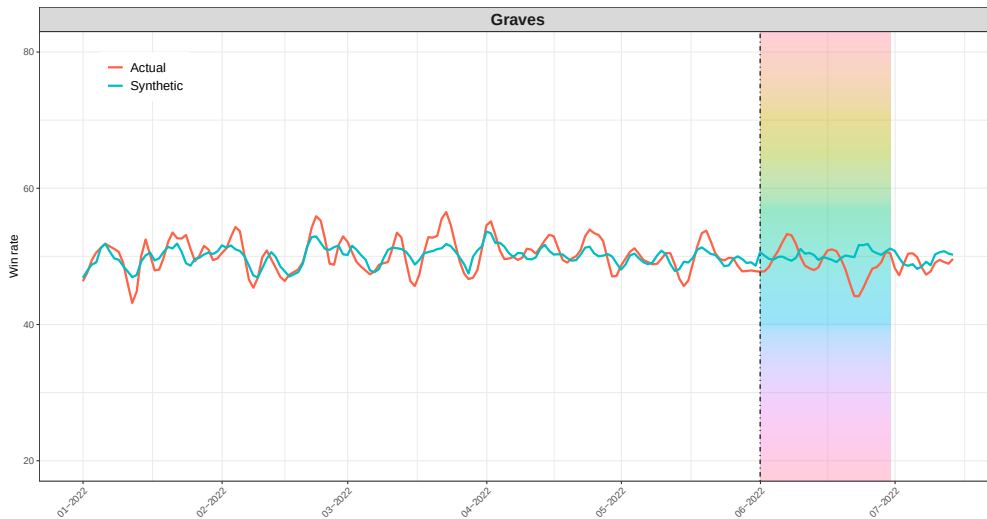


Figure 5: Graves’ daily win rates and synthetic control estimation results. The actual series is smoothed by a Nadaraya-Watson regression before applying the synthetic control estimator. The dashed vertical line denotes the day of disclosure, and the rainbow area highlights LGBT Pride Month.

³³ Examples of such websites include <https://lolalytics.com/lol/graves/build/> and <https://www.leagueofgraphs.com/champions/stats/graves>.

that no game-relevant skills or attributes were altered during the treatment period, and they could observe that Graves’s strength remained consistent. These factors suggest that the negative impact of the coming-out event estimated in Section 4 is unlikely to be driven by actual or presumed changes in character relative strengths.

5.2 Players’ Skills

If highly skilled players exhibit distinct preferences for Graves or are less influenced by the character’s sexual orientation, the decision to switch from Graves might be driven by gameplay factors rather than social preferences for sexual orientation, thus challenging our social stigma narrative.

To address this concern, we examine the correlation between players’ skills and their decision to abandon Graves. We classify players into two groups based on their preferences for Graves before his disclosure: the first group comprises those who chose Graves in at least 5% of their matches before the coming-out event (henceforth labeled as *prior users*, $n = 953$), while the second group comprises the remaining players (henceforth labeled as *non-prior users*, $n = 6468$). We then examine performance differences both within and between these groups before and after the treatment.

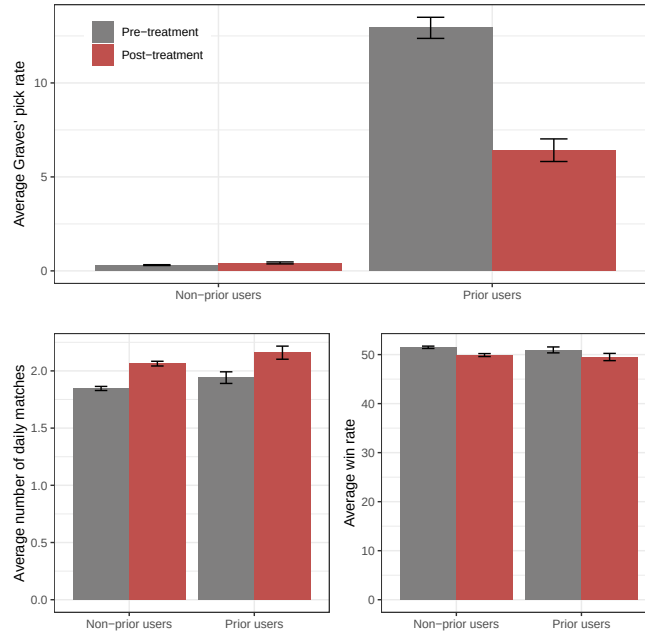


Figure 6: Players’ average pick rates for Graves and performance measures. Players are divided into two groups based on their preferences for Graves before his disclosure. The panels display the average pick rates for Graves, number of daily matches, and win rate of each group before and after the coming-out event.

The top panel of Figure 6 displays the average pick rate for Graves among prior and non-prior users before and after the treatment. We observe a sharp decline in pick rates among prior users following the coming-out event, similar in size to the decrease shown in Figure 2 and Table 1. Conversely, non-prior users exhibit a marginal increase in average pick rates post-treatment, although this increase is practically negligible.

In the remaining panels of Figure 6, we investigate whether prior and non-prior users differ in their in-game behavior. First, the bottom left panel shows the average number of daily matches played by each group. We observe similar engagement levels across groups, both before and after the treatment, indicating that prior and non-prior users tend to play a comparable number of matches per day. Notably, neither group appears to disengage from the game following Graves’ disclosure. On the contrary, the number of daily matches slightly increases post-treatment for both groups, suggesting that players are not leaving the game but instead redirecting their attention to other characters. Supporting this interpretation, Figure A.VI in Appendix A shows no meaningful change in role preferences for either group following the event. This pattern is consistent with the idea that players face low switching costs and have access to suitable alternatives, enabling them to express their preferences without disrupting their gameplay. These results align with those displayed in Table 1.

Second, the bottom right panel displays the players’ average win rates. We observe no substantial disparities within and between groups, indicating that the preference for Graves and the decision to abandon this character are unrelated to players’ skill levels. Overall, these findings dismiss the possibility that game-play factors are the driving force behind the estimated effects of Section 4, lending additional support to the social stigma attached to playing an LGB character as the mechanism underlying the players’ observed behavior.

5.3 Players’ Performance

To ensure the accuracy of our measurement of players’ genuine attitudes toward the LGB community, it is crucial to assess whether shifting away from Graves to other characters impacts players’ performance. If there are performance costs, our estimates could be biased toward zero, as players might continue using Graves for strategic considerations. Moreover, if players switch characters primarily for convenience, our analysis might unintentionally capture a different phenomenon instead of the intended social stigma.

We employ difference-in-differences identification and estimation strategies to assess the

impact of players abandoning Graves on their performance. We gauge players' performance by their daily win rate. Our analysis focuses on the 953 prior-users of Section 5.2, who are classified into treated or control groups based on their responses to Graves' disclosure. We consider two different definitions of the treatment, sorted by their intensity. In the first version, labeled *moderate reduction*, we classify as treated those players who decreased their average pick rate for Graves following his disclosure by at most 75% of their pre-treatment average pick rate (the number of treated units is 287). In the second version, labeled *substantial reduction*, we classify as treated those players who reduced their average pick rate for Graves by 75% to 100% post-disclosure (the number of treated units is 491).³⁴ In both scenarios, the control group remains the same and consists of the 175 prior users who did not reduce their average pick rate for Graves following his disclosure.

Under the standard assumptions of parallel trends and no anticipation (see, e.g., Roth et al., 2023), we can identify the average treatment effect on the treated (ATT) using observable data. The parallel trend assumption posits that the performance of treated and untreated players would have evolved similarly if Graves' disclosure had not occurred. While we cannot formally test this assumption, the findings of Section 5.1 and Section 5.2 provide substantial support for its plausibility.³⁵ As for the no anticipation assumption, it stipulates that in the weeks preceding the disclosure, players' performance did not change due to the incoming Graves' disclosure. The plausibility of this assumption was thoroughly discussed in Section 2.2 and Section 4.1.

We implement the approach of Callaway and Sant'Anna (2021) to target the ATT at a particular day $t > T^{pre}$:³⁶

$$ATT(t) := \mathbb{E}[Y_{i,t}(1) - Y_{i,t}(0) | D_i = 1], \quad (6)$$

where potential outcomes are defined as in Section 3.2, and D_i is a binary variable indicating whether a player is treated or not. Under the assumptions of parallel trends and no anticipation,

³⁴ Figure A.VII in Appendix A shows the distribution of percentage reduction in average pick rates for Graves from pre-treatment to post-treatment among prior users.

³⁵ Moreover, we demonstrate below the absence of pre-treatment differences in trends by reporting placebo estimates of the ATT that are not statistically different from zero. This is often viewed as a natural plausibility check, although even if pre-trends are perfectly parallel, this does not necessarily guarantee the satisfaction of the post-treatment parallel trends assumption (see, e.g., Roth et al., 2023).

³⁶ The framework outlined in Callaway and Sant'Anna (2021) is broader as it accommodates multiple groups defined by the timing of treatment reception. This enables the identification and estimation of the group-time ATTs, defined as $ATT(g, t) := \mathbb{E}[Y_{i,t}(g) - Y_{i,t}(0) | G_g = 1]$, where G_g is a binary variable indicating treatment reception in period g . However, our data set features a single group, given that all treated players receive the treatment simultaneously (i.e., at Graves' disclosure date). This allows us to simplify notation and focus on the time ATTs in equation (6) for the single group we observe.

Callaway and Sant’Anna (2021) show that $ATT(t)$ can be identified by comparing the change in outcomes between the latest period before the coming-out event and day t experienced by treated players to the change in outcomes experienced by control players.³⁷

Figure 7 displays the point estimates and simultaneous 95% confidence bands for the $ATT(t)$. Overall, we find that shifting away from Graves to other characters has no impact on players’ performance. None of the estimated $ATT(t)$ is statistically different from zero, suggesting that transitioning to other characters does not result in any performance-related consequences. This finding highlights that the decision to move away from Graves is not influenced by performance considerations.

Figure 7 further displays placebo estimates of the time ATTs for the ten days before the treatment.³⁸ As explained above, these estimates are valuable for “pre-testing” the credibility of the parallel trend assumption (Callaway & Sant’Anna, 2021). Notably, all placebo time ATTs in the pre-treatment periods are statistically insignificant, supporting the validity of the parallel trends assumption.

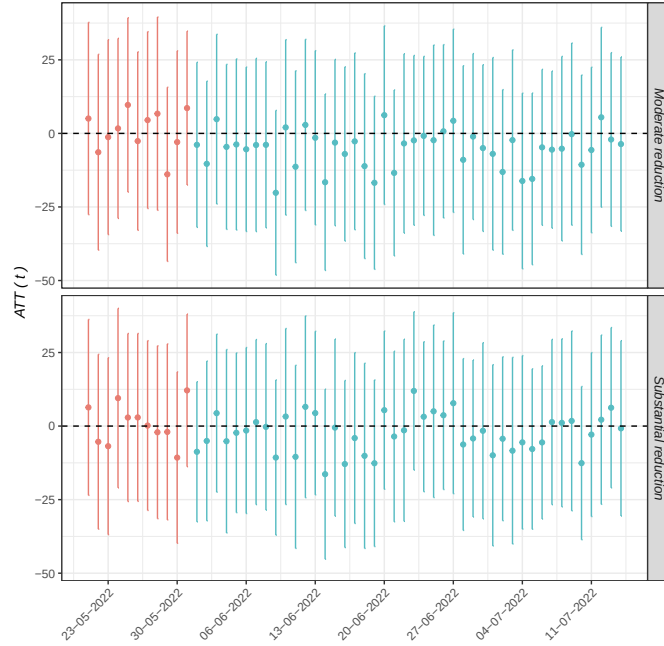


Figure 7: Point estimates and simultaneous 95% confidence bands allowing for clustering at the player level for the $ATT(t)$. Red lines refer to pre-treatment periods, while blue lines refer to post-treatment periods. Each row corresponds to a different version of the treatment.

³⁷ Formally, Callaway and Sant’Anna (2021) show that $ATT(t) = \mathbb{E}[Y_{i,t} - Y_{i,Tpre} | D_i = 1] - \mathbb{E}[Y_{i,t} - Y_{i,Tpre} | D_i = 0]$. Estimation is carried out by replacing expectations with their sample analogs.

³⁸ Figure A.VIII in Appendix A displays the remaining estimated placebo $ATT(t)$.

5.4 New Substitute Character

On June 9th, 2022, a new character (*Bel’Veth*) was released.³⁹ Since the primary position the character is designed for is the same position as Graves is designed for, it can be considered a close substitute. Therefore, players’ decisions to switch away from Graves (after June 9th) might to some degree be driven by the desire to experiment with the new character and to explore potential competitive advantages, challenging social stigma as the primary explanation behind our main result.

If the release serves as the primary explanation for the observed drop in Graves pick rate we would expect a positive correlation between the size of the drop in players’ Graves usage and players preferences for the new character. In Figure 8 we show that this is not the case. Figure 8 shows the average pick rate for *Bel’Veth* for the prior users of Section 5.2, classified as in Section 5.3. We find that those players who feature a substantial reduction in their Graves’ pick rates following the coming-out event are less likely to pick *Bel’Veth* for their matches than players who feature a moderate reduction. Moreover, these players are even less likely to pick *Bel’Veth* for their matches than players who did not react at all to Graves’ disclosure. We therefore argue that the release is unlikely to serve as the primary explanation for the observed main result of the paper.

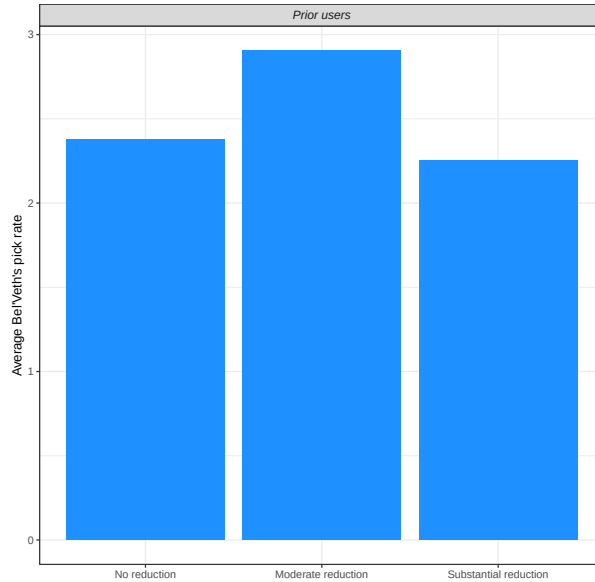


Figure 8: Average pick rate for *Bel’Veth* after its release among prior users.

³⁹ Figure A.IX in Appendix A displays *Bel’Veth*’s daily pick rates.

5.5 Coming Out versus LGBT Pride Month

As described in Section 2.2, the disclosure of Graves’ sexual orientation coincided with the start of LGBT Pride Month. This means that the coming-out event encompasses two “simultaneous treatments” (see, e.g., Roller & Steinberg, 2023), namely the announcement of Graves’ homosexuality and the introduction of visual and expressive elements in League of Legends that support the LGBT community. It is therefore plausible that the findings presented in Section 4 may, to some extent, be influenced by the presence of LGBT Pride Month, which might elicit negative reactions from certain players, leading them to shift their preferences away from LGB characters. While this alternative perspective does not undermine the validity of our identification strategy, it does raise questions about our interpretation of the estimated effects as solely stemming from Graves’ disclosure.

In Appendix D, we introduce a theoretical framework that formalizes the existence of two simultaneous treatments and discuss the implications for interpretation. Additionally, we outline sufficient assumptions that enable us to separate the impacts of coming out and LGBT Pride Month on players’ preferences for Graves. Here, we provide the main intuitions behind our approach, directing the reader to the appendix for technical details.

To examine the potential impact of LGBT Pride Month on players’ preferences for LGB characters, we leverage the existence in our data set of other four characters (*Diana*, *Leona*, *Nami*, and *Neeko*) already acknowledged as part of the LGB community before the coming-out event. These characters are subject only to a part of our treatment, specifically being part of the LGB community while LGBT Pride Month is ongoing, whereas Graves experiences both the disclosure of his sexual orientation and LGBT Pride Month.

We create a composite LGB unit by averaging the pick rates of Diana, Leona, Nami, and Neeko and employ the synthetic control estimator described in Section 3.2 to estimate the effect of LGBT Pride Month on players’ preferences for LGB characters. Then, under the assumption that the influence of LGBT Pride Month is uniform across all LGB characters, we can compare the results with those obtained for Graves to separate the impacts of coming out and LGBT Pride Month on players’ preferences for Graves. Intuitively, if the estimated impact of LGBT Pride Month on players’ preferences for LGB characters is small relative to the estimated impact of the coming-out event on players’ preferences for Graves, this suggests that the findings of Section 4 must be primarily attributed to Graves’ disclosure.

Figure 9 displays the actual and the synthetic pick rate series for the composite LGB unit.

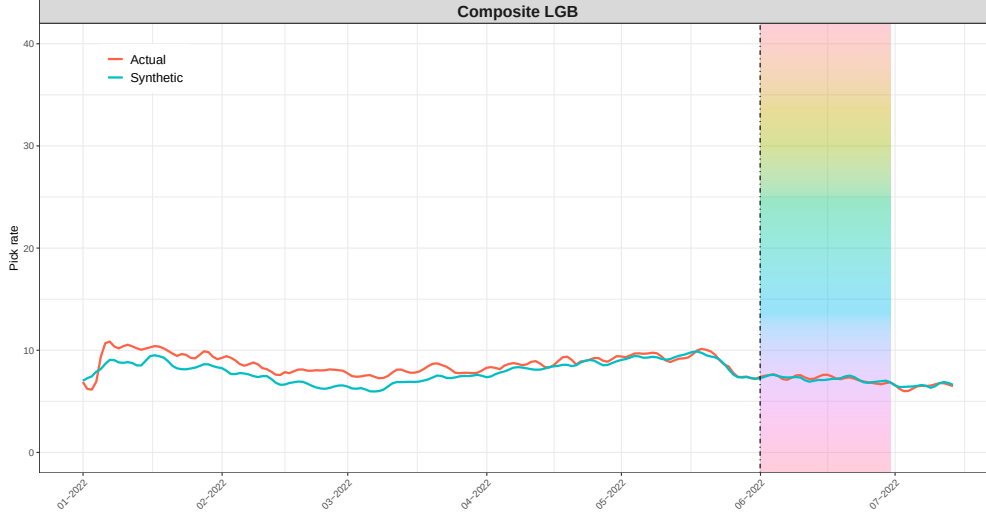


Figure 9: Composite LGB unit's daily pick rates and synthetic control estimation results. The actual series is smoothed by a Nadaraya-Watson regression before applying the synthetic control estimator. The dashed vertical line denotes the day of disclosure, and the rainbow area highlights LGBT Pride Month.

Overall, our analysis suggests that LGBT Pride Month had no impact on players' preferences for LGB characters. Before the treatment, the synthetic control estimator closely aligns with the actual series, providing support for the estimator's ability to predict the counterfactual series. After the treatment date, the synthetic control estimator continues to align with the actual series, confirming that the players' preferences for LGB characters were unaffected by LGBT Pride Month. The average effect is estimated to be -0.215 percentage points (standard error: 3.052), and the conventional 95% confidence interval encompasses zero, indicating a failure to reject the null hypothesis of no effect. Under the homogeneity assumption discussed above, these findings support the interpretation that the estimated effects presented in Section 4 are primarily driven by Graves' disclosure rather than being influenced by the broader context of LGBT Pride Month.

6 Conclusion

Discrimination based on sexual orientation is first and foremost a human rights issue. However, when individuals with a stigmatized identity are unfairly targeted in education, health, social, and political settings, there is a loss of human capital that can have detrimental effects on the economy as a whole Badgett, 2020. For example, bullying and discrimination act as barriers to students' acquisition of skills and knowledge. Furthermore, even short experiences of bullying can have severe long-term health consequences Boden et al., 2016. Therefore, understanding

the barriers that individuals from stigmatized groups face is of large societal importance.

In this study, we utilize a comprehensive data set sourced from the widely popular online video game *League of Legends* and exploit exogenous variation in the identity of a playable character to credibly identify sentiments towards LGB status. Players in the game select a playable character before each match. Each playable character is characterized by game-relevant attributes and a background story. Leveraging an unexpected revelation during the 2022 LGBT Pride Month, wherein game developers disclosed the sexual orientation minority status of a playable character, we investigate individuals' responses to this disclosure. By tracking players' revealed preferences for the character over a meaningful period, we provide insights into reactions following the disclosure. To isolate the effects on player preferences from potential confounding influences, we employ synthetic control methods. Our findings reveal a substantial and persistent negative impact, with preferences for this character decreasing by more than 30% over a meaningful period. This underscores the potential negative consequences of disclosing one's sexual minority status and provides a rationale for the underrepresentation of individuals with LGB status in many regions and professions.

To bolster the credibility of stigma as the primary explanation for the estimated effects, we address and eliminate several alternative channels. First, we rule out the possibility that shifts in characters' relative strengths could explain our estimated effect. Second, we show that players' skills have no correlation with the choice to drop the character. Third, we provide evidence that switching to other characters does not affect the performance of the players involved and that the release of new characters in the post-treatment period is unlikely to serve as primary explanation behind the results. Fourth, we introduce a theoretical framework that formalizes the existence of two "simultaneous treatments" and use information on other playable characters that belong to the LGB group to rule out that LGBT Pride Month serves as the main explanation for the observed result.

We have dismissed the possibility that any actual or presumed change in the character's strength is driving our results. This strongly suggests that the estimated cost of coming out is unlikely to be driven by factors other than stigmatization. This insight holds significant implications for policymakers aiming to develop interventions that effectively tackle stigmatization and improve the overall well-being of LGB individuals. Policies should be formulated to discourage discriminatory behavior, either by increasing its costs or by creating inclusive social environments that promote the acceptance of sexual minority individuals and reduce the stigma.

Raising awareness about the reaction to sexual minority disclosure could be an important step to develop such a society, as previous research has demonstrated the value of discrimination awareness for minority groups outcomes Pope et al., [2018](#). At the same time, policymakers may also consider providing resources and support to individuals who have recently come out, such as access to counseling and mental health services. By doing so, they can mitigate some of the negative outcomes that may arise from coming out.

Appendix A Additional Figures and Tables

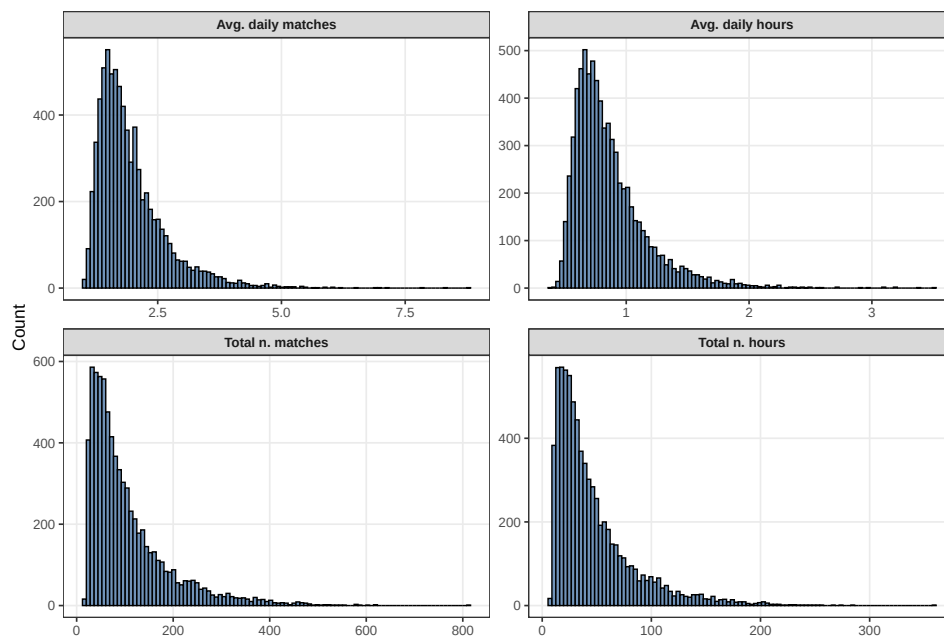


Figure A.I: Histograms of players' total and average daily activity, measured in both number of matches played and hours spent in-game.

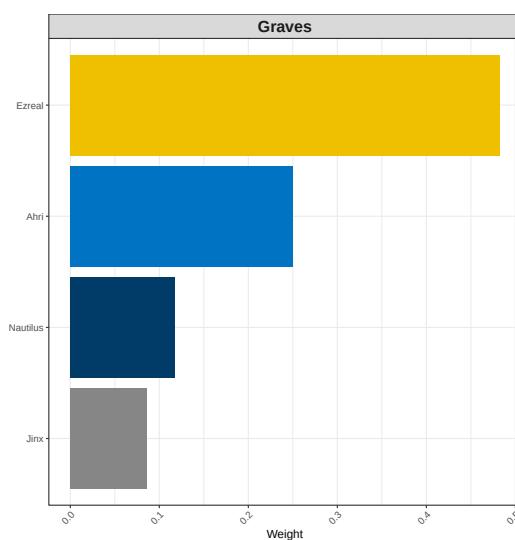


Figure A.II: Identities and contributions of characters in the donor pool for the Graves' synthetic control displayed in Figure 2.

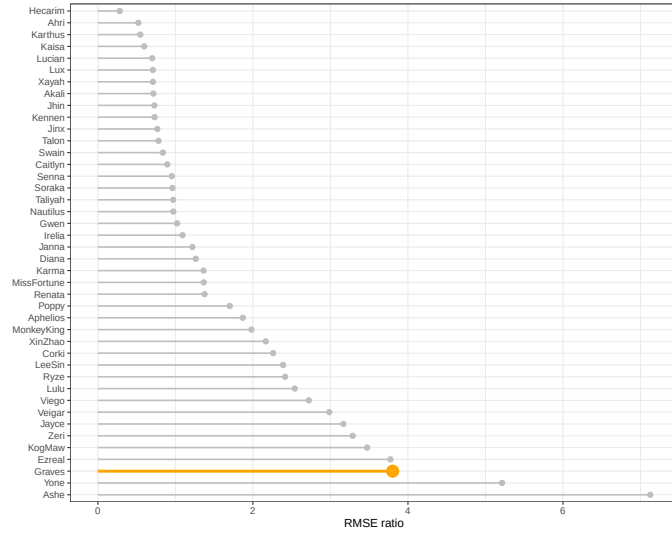


Figure A.III: Ratio of post- to pre-treatment RMSE for Graves and control characters. Only control characters with a pre-treatment RMSE of at least 1 are considered to exclude overfit specifications.

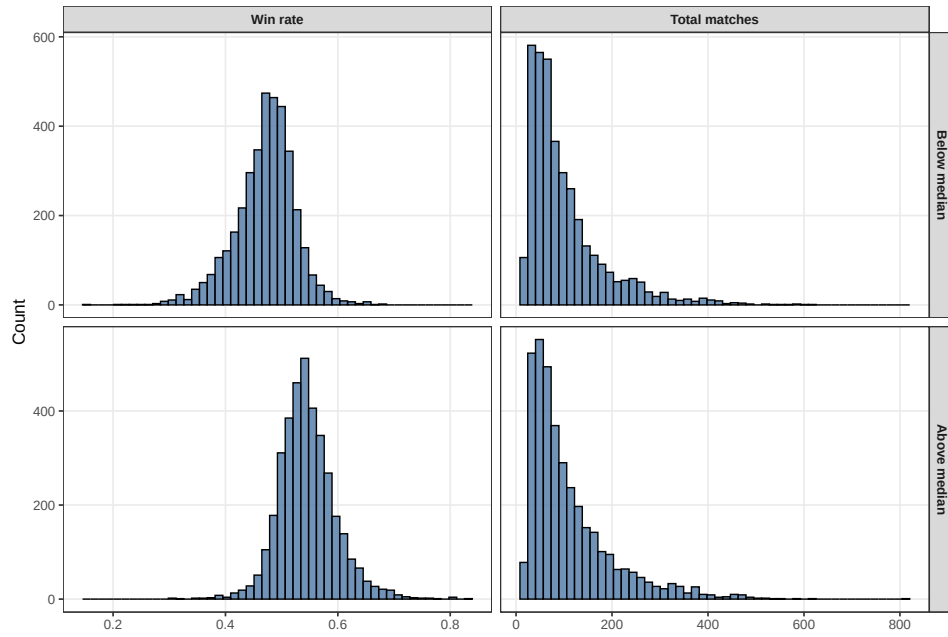


Figure A.IV: Distribution of overall win rates and total matches played for players classified as above or below the median pre-treatment win rate.

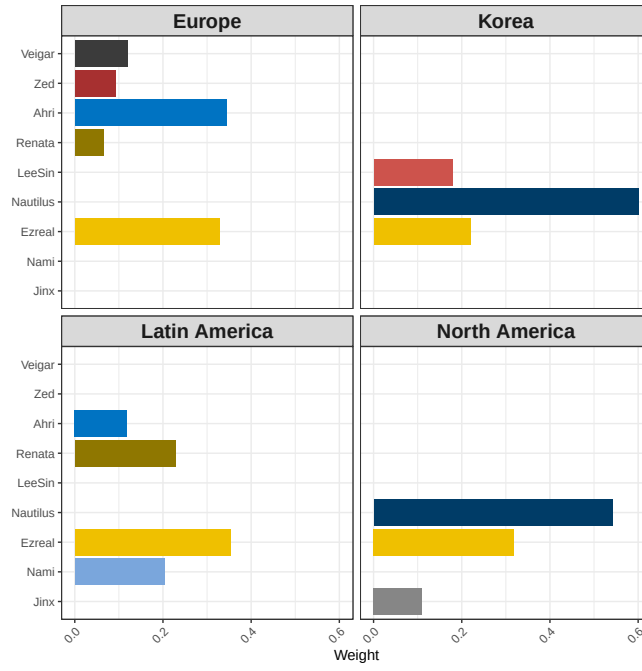


Figure A.V: Identities and contributions of characters in the donor pool for the Graves' synthetic controls displayed in Figure 4.

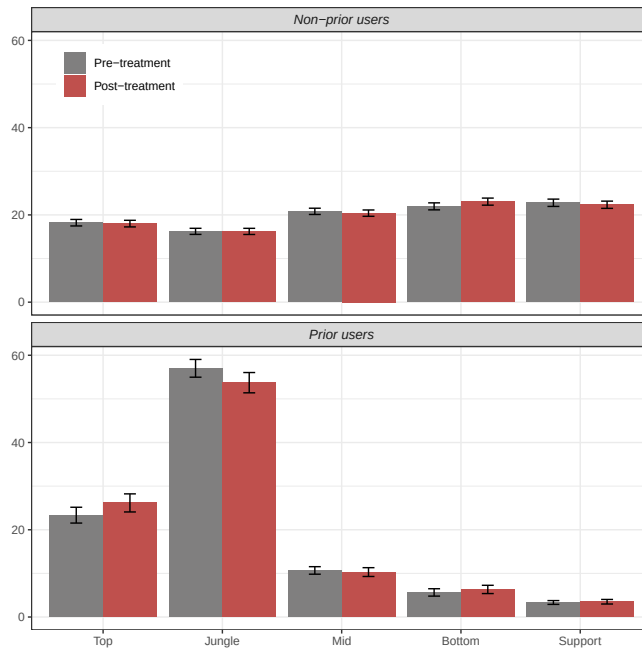


Figure A.VI: Shares of matches played in each role before and after the coming-out event. Players are divided into two groups based on their preferences for Graves before his disclosure.

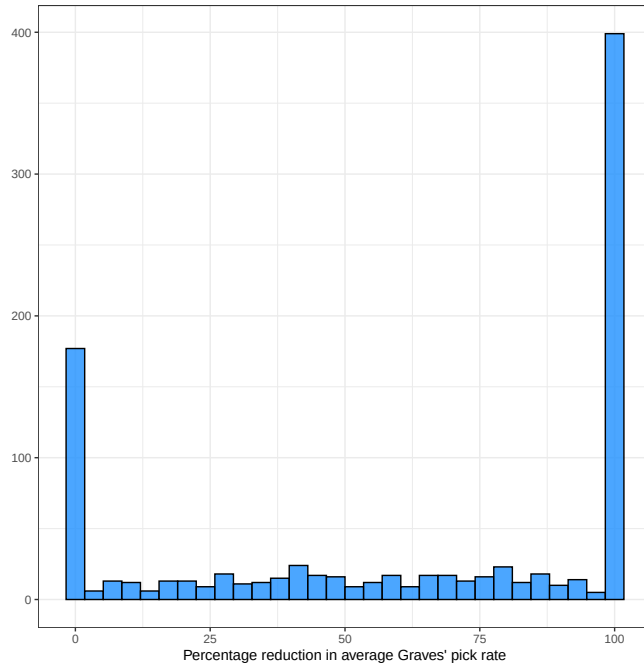


Figure A.VII: Distribution of percentage reduction in average pick rates for Graves from pre-treatment to post-treatment among prior users.

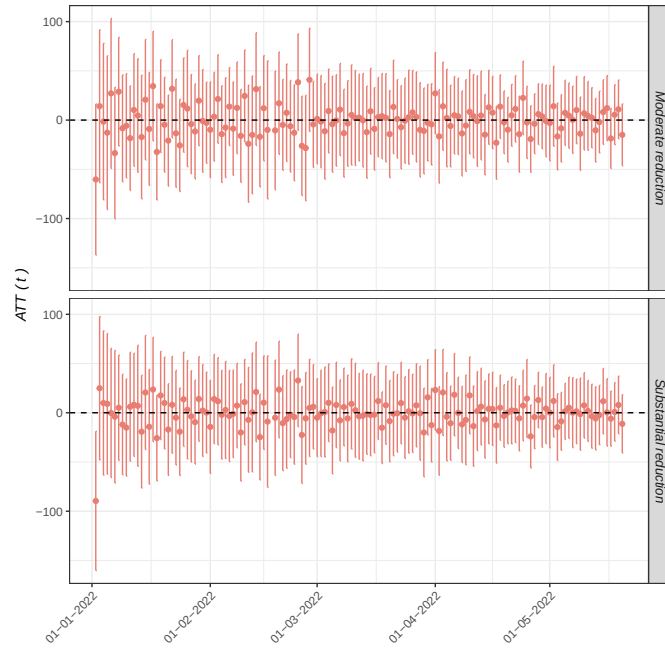


Figure A.VIII: Point estimates and simultaneous 95% confidence bands allowing for clustering at the player level for the placebo $ATT(t)$. Each row corresponds to a different version of the treatment.

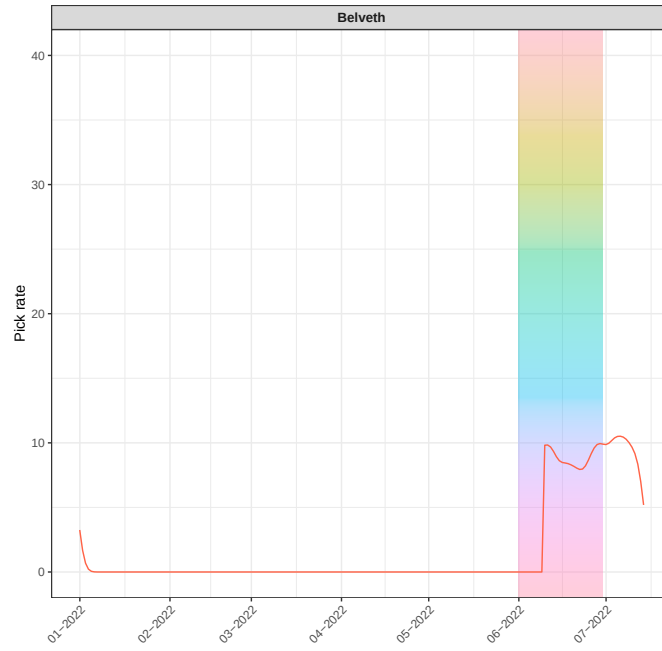


Figure A.IX: Bel’Veth’s daily pick rates. The series is smoothed by a Nadaraya-Watson regression. The rainbow area highlights LGBT Pride Month. The non-zero pick rate displayed in early January is a result of the smoothing.

Table A.I: Most popular characters.

	<i>Pre-treatment</i>					<i>Post-treatment</i>				
	<i>Top</i>	<i>Jungle</i>	<i>Mid</i>	<i>Bottom</i>	<i>Support</i>	<i>Top</i>	<i>Jungle</i>	<i>Mid</i>	<i>Bottom</i>	<i>Support</i>
1	Irelia (12.007)	LeeSin (19.132)	Ahri (13.62)	Jinx (24.5)	Karma (19.522)	Gangplank (12.455)	Viego (19.419)	Yone (17.587)	Ezreal (29.02)	Renata (16.43)
2	Camille (10.4)	Graves (18.557)	Akali (12.816)	Ezreal (21.155)	Nautilus (15.302)	Fiora (10.569)	MonkeyKing (15.064)	Sylas (13.177)	Zeri (19.716)	Karma (16.239)
3	Jayce (9.756)	Viego (17.982)	Yone (12.197)	Jhin (20.746)	Lulu (14.322)	Aatrox (9.451)	LeeSin (12.78)	Ahri (12.83)	Twitch (14.751)	Yuumi (14.215)
4	Fiora (8.777)	Diana (13.771)	Yasuo (10.611)	Kaisa (17.174)	Nami (11.781)	Irelia (9.407)	Taliyah (12.322)	Yasuo (10.855)	Jhin (14.531)	Senna (13.715)
5	Aatrox (8.616)	Khazix (9.448)	Viktor (10.525)	Lucian (12.505)	Pyke (11.401)	Kayle (8.51)	Graves (12.319)	Viktor (10.347)	Kaisa (13.317)	Lulu (13.274)

Notes. Most popular characters by role based on average pick rates (displayed in parenthesis)

Table A.II: Regional results.

	<i>Synthetic Controls</i>		<i>Regularized Synthetic Controls</i>	
	(1) All characters	(2) Only non-substitutes	(3) All characters	(4) Only non-substitutes
Panel 1: <i>Europe</i>				
$\hat{\tau}$	-9.062	-11.719	-8.696	-10.670
95% CI	[-15.832, -2.292]	[-19.479, -3.958]	[-14.210, -3.183]	[-19.594, -1.758]
N. Donors	5	5	6	8
RMSE	2.206	2.639	2.155	2.583
Pre-treatment average	14.618	14.618	14.618	14.618
Panel 2: <i>Korea</i>				
$\hat{\tau}$	-10.015	-8.955	-10.299	-8.990
95% CI	[-18.262, -1.768]	[-20.502, 2.593]	[-17.720, -2.878]	[-18.368, 0.376]
N. Donors	3	3	3	3
RMSE	6.211	6.550	6.249	6.594
Pre-treatment average	31.441	31.441	31.441	31.441
Panel 3: <i>Latin America</i>				
$\hat{\tau}$	-6.212	-5.899	-5.206	-4.960
95% CI	[-11.850, -0.574]	[-12.449, 0.651]	[-10.645, 0.233]	[-12.531, 2.605]
N. Donors	4	4	7	6
RMSE	2.575	2.552	2.514	2.522
Pre-treatment average	16.217	16.217	16.217	16.217
Panel 4: <i>North America</i>				
$\hat{\tau}$	0.939	0.940	1.110	1.130
95% CI	[-4.583, 6.462]	[-6.259, 8.138]	[-3.412, 5.632]	[-5.807, 8.079]
N. Donors	3	3	4	4
RMSE	4.669	4.669	4.681	4.685
Pre-treatment average	22.282	22.282	22.282	22.282

Notes. Point estimates and 95% confidence intervals for $\hat{\tau}$. Additionally, the number of donors receiving a non-zero weight and the pre-treatment root mean squared error are displayed. Each panel reports the results obtained using only matches from a particular macro-region. Each column corresponds to a different specification, with the specifications differing solely in the employed estimator and donor pool composition.

Appendix B Robustness Checks

This section presents a series of robustness checks that support the credibility of our findings. We begin with checks related to our main results, followed by additional analyses that validate the player heterogeneity findings.

B.1 Main Results

We examine the robustness of our main results from Section 4.1 to alternative estimation strategies and variations in the composition of the donor pool. In particular, we repeat our analysis employing a regularized synthetic control estimator (see Section 3.2) and explore different donor pool configurations focusing on characters from distinct roles. Notably, Graves is predominantly

designed for and played in three of the possible roles within a team. Consequently, there is a possibility of spillover effects on other characters mainly played in these positions, as players transitioning away from Graves are likely to switch to these alternatives.⁴⁰ To mitigate this potential for spillover effects, we restrict our donor pool to characters that are “non-substitutes” of Graves, that is, those primarily designed for the remaining two roles.⁴¹

Table B.I displays the results. For any donor pool composition, the results are not sensitive to the choice of the regularization parameter, and point estimates are consistently negative. The results are consistent also quantitatively across all specifications, with a decline ranging between 40.98% and 38.17% of the pre-treatment average preferences for Graves. Overall, these results support our main finding of a substantial negative impact of the coming-out event on players’ preferences for Graves.

We also assess the credibility of the synthetic control estimator by conducting a robustness check that artificially shifts the coming-out event ten days earlier. This backdating exercise allows us to evaluate the estimator’s predictive accuracy during a ten-day hold-out period (see e.g., Abadie & Vives-i-Bastida, 2022). The upper panel of Figure B.I presents the results of this analysis. We observe three key findings. First, the estimated effects remain qualitatively and quantitatively consistent, confirming a negative and persistent impact of the coming-out event on players’ revealed preferences for Graves. Second, the synthetic control estimator demonstrates a good fit during the hold-out period, indicating its ability to accurately capture Graves’ behavior prior to the disclosure. Third, the actual and the synthetic series begin to diverge on

Table B.I: Main results.

	<i>Synthetic Controls</i>		<i>Regularized Synthetic Controls</i>	
	(1) All characters	(2) Only non-substitutes	(3) All characters	(4) Only non-substitutes
$\hat{\tau}$	-7.606	-7.161	-7.323	-7.080
95% CI	[-13.542, -1.670]	[-15.383, 1.062]	[-11.471, -3.174]	[-14.831, 0.665]
N. Donors	4	4	4	5
RMSE	2.365	2.525	2.330	2.388
Pre-treatment average	18.558	18.558	18.558	18.558

Notes. Point estimates and 95% confidence intervals for $\hat{\tau}$. Additionally, the number of donors receiving a non-zero weight and the pre-treatment root mean squared error are displayed. Each column corresponds to a different specification, with the specifications differing solely in the employed estimator and donor pool composition.

⁴⁰ Table 1 and the findings of Section 5.2 support this intuition.

⁴¹ Graves is predominantly designed for and played in the *top*, *jungle*, and *mid* positions. Therefore, we consider characters designed for and played in the *bottom* and *support* positions as “non-substitutes.”

the true day of disclosure, even when the estimator has no knowledge of the actual disclosure date. The absence of estimated effects before the coming-out event also lends support to the plausibility of a no-anticipation assumption (see e.g., Abadie, 2021).

Finally, we conduct an additional robustness test by performing a leave-one-out exercise, where we repeatedly estimate the synthetic control series by excluding one character with non-zero estimated weights at a time from the donor pool (see e.g., Abadie, 2021). The lower panel of Figure B.I presents the results of this analysis. Overall, our finding of a negative and persistent impact of the coming-out event on players' preferences for Graves is robust to the exclusion of any particular character. Most of the leave-one-out synthetic series closely align with the main estimate, thus reinforcing the robustness of the main conclusion of our study. One leave-one-out series falls beneath the other synthetic series, suggesting a somewhat reduced, although still negative, impact. However, this series diverges from the actual series in the weeks prior to the treatment, which undermines the reliability of its results.⁴²

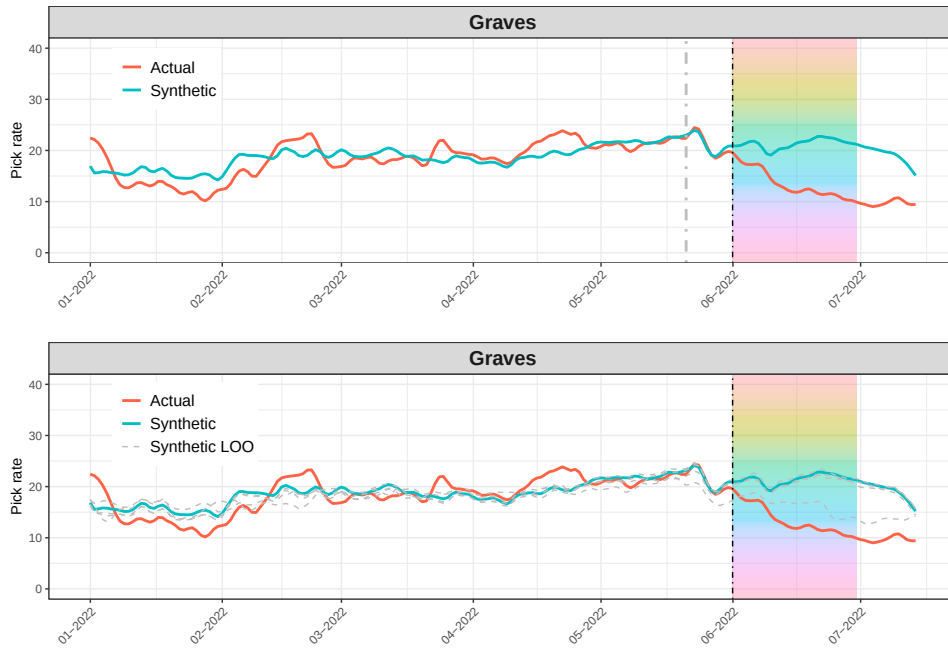


Figure B.I: Robustness checks for main results. The upper panel shifts the coming-out event ten days earlier, with the new treatment date denoted by the vertical gray dashed line. The lower panel reports leave-one-out estimates of the synthetic control series, obtained by excluding one of the characters of Figure A.II at a time from the donor pool.

⁴² This series is obtained by excluding the character *Ezreal* from the donor pool. Figure A.II in Appendix A shows that this character receives the largest weight in our main specification. Therefore, the divergence of this leave-one-out series from the actual series is unsurprising.

B.2 Player Heterogeneity

We assess the robustness of our player heterogeneity results from Section 4.2 using the same backdating and leave-one-out exercises described in the previous subsection. Figure B.II displays the results. For both skill groups, we observe that even when the estimator has no knowledge of the actual disclosure date, the actual and synthetic series begin to diverge precisely on the true day of disclosure. The fit during the hold-out period is also strong, indicating good predictive performance. Moreover, most of the leave-one-out synthetic series closely track the main estimates, reinforcing the robustness of our findings.

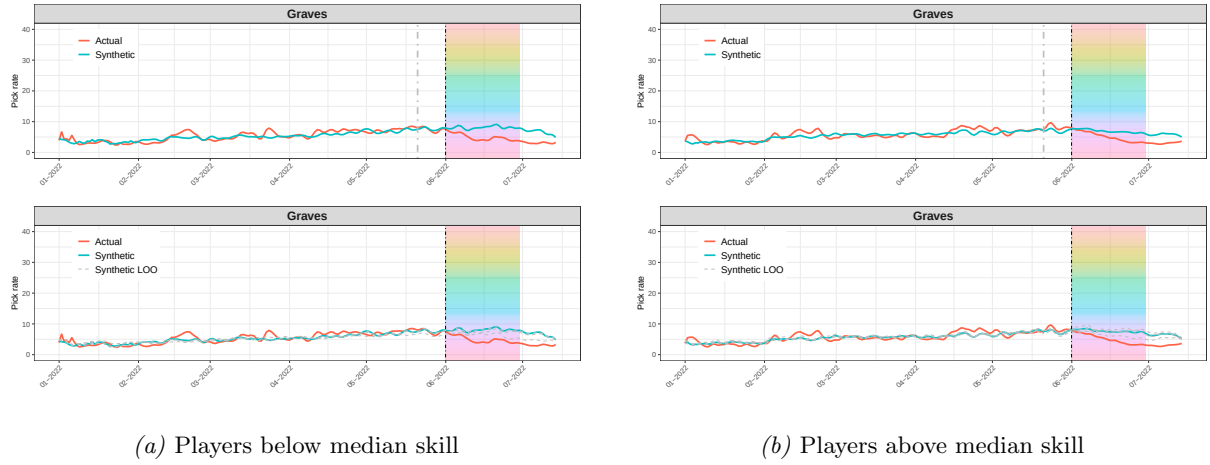


Figure B.II: Robustness checks for player heterogeneity results. The upper panels shift the coming-out event ten days earlier, with the new treatment date denoted by the vertical gray dashed line. The lower panels report leave-one-out estimates of the synthetic control series, obtained by excluding one character with non-zero estimated weights at a time from the donor pool.

Appendix C Twisted Fate

In the League of Legends universe, Graves, described as a “gruff-looking, broad-shouldered, middle-aged man,” forms a criminal partnership with Twisted Fate, who is characterized as “a tall, handsome male with tanned skin, trimmed beard, and long dark hair.” Together, they engage in various illicit activities until, in the midst of a heist, Graves finds himself captured. This event leads Graves to perceive a betrayal by Twisted Fate, prompting a pursuit of revenge upon his escape. However, the duo ultimately decides to reconcile their differences and resumes their collaboration.⁴³

Notably, one of the initially considered narrative concepts for Graves and Twisted Fate

⁴³ The complete background of Graves and Twisted Fate is available at <https://leagueoflegends.fandom.com/wiki/Graves> and https://leagueoflegends.fandom.com/wiki/Twisted_Fate.

involved them being married or ex-lovers. Although this particular aspect was discarded, the general narrative retained the notion of “palpable sexual tension” between the two characters.

The story unveiling Graves’ sexual orientation (see Section 2.2) also subtly hints at Twisted Fate’s pansexuality, although this is not explicitly stated. Perhaps, the most notable passage that alludes to this is:

No matter the size, shape, make, or model, none can resist the charms of Tobias Felix. I have conned hundreds—nay, thousands—of dew-eyed tourists across the whole of this vast and gullible land. (Twisted Fate)

We investigate whether this implied revelation has captured the players’ attention in Figure C.I, illustrating the Google search interest for the queries “Graves gay” and “Twisted Fate gay.” Throughout 2022, we observe approximately no interest in the latter query, with a small spike occurring during the week of the coming-out event, amounting to less than half of the spike associated with Graves. Furthermore, the search interest for Twisted Fate is always lower than that for Graves, suggesting the greater popularity of Graves among players.⁴⁴ These results underscore the relatively low attention directed towards Twisted Fate from players, who were primarily focused on Graves and the explicit establishment of his sexual orientation. As a result, we concentrate our analysis on Graves and his disclosure for a more credible identification of the effects of coming out.

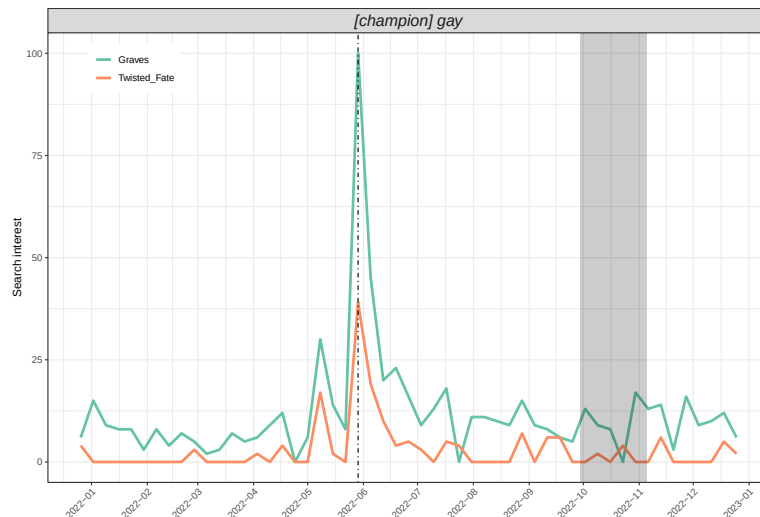


Figure C.I: Google search interest over time for the queries “Graves gay” and “Twisted Fate gay.” The dashed vertical line denotes the week of disclosure, and the shaded area highlights the League of Legends World Championship.

⁴⁴ This is also suggested by Graves’s pick rates being approximately 3 to 4 times higher than those of Twisted Fate, as displayed in Figure C.II.

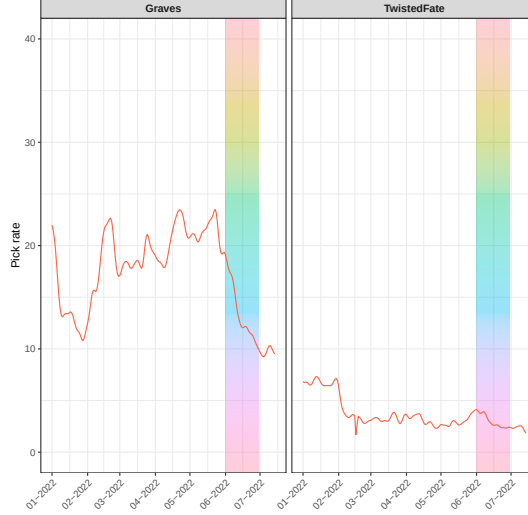


Figure C.II: Graves and Twisted Fate’s daily pick rates. The series are smoothed by a Nadaraya-Watson regression. The rainbow area highlights LGBT Pride Month.

Appendix D Anatomy of the Coming-Out Event

In this section, we discuss how the existence of two treatments - the disclosure of Graves’ sexual orientation and the start of LGBT Pride Month - occurring at the same time may affect the interpretation of the main findings of Section 4.1. The notation follows that used in Section 3.2. The results of the analysis are detailed in Section 5.5.

In the next subsection, we introduce the framework that formalizes the existence of two “simultaneous treatments.” We then outline sufficient assumptions that enable us to separate the impacts of coming out and LGBT Pride Month on players’ preferences for Graves.

D.1 Simultaneous Treatments

As described in Section 2.2, the disclosure of Graves’ sexual orientation coincided with the start of LGBT Pride Month. This means that the coming-out event encompasses two treatments occurring at the same time, namely the announcement of Graves’ homosexuality and the introduction of visual and expressive elements in League of Legends that support the LGBT community.⁴⁵

We recognize the potential influence of LGBT Pride Month on players’ preferences for characters by introducing the binary variable $L_i \in \{0, 1\}$ to represent character i ’s inclusion in the LGB community no later than $T^{pre} + 1$. Consequently, we observe three distinct groups of

⁴⁵ See, e.g., Roller and Steinberg (2023) for a discussion on “simultaneous treatments” and methodologies for disentangling their effects under a Difference-in-Differences identification strategy.

units: the first group includes only Graves, with $C_i = L_i = 1$; the second group includes only Diana, Leona, Nami, and Neeko, with $C_i = 0$ and $L_i = 1$; and the third group includes all other characters, with $C_i = L_i = 0$.⁴⁶

To explicitly account for the influence of the two treatments C_i and L_i , we define the potential pick rates as $Y_{i,t}^{C,L}$. Then, for each period $t > T^{Pre}$, the effect of the coming-out event on players' preferences for Graves in (1) corresponds to:

$$\tau_t = Y_{1,t}^{1,1} - Y_{1,t}^{0,0} \quad (\text{D.1})$$

Equation (D.1) shows why we need to be cautious in interpreting the estimated effects of Section 4.1 as solely stemming from the disclosure of Graves' sexual orientation. Under an extended version of the SUTVA assumption (see Section D.2), we observe $Y_{1,t} = Y_{1,t}^{1,1}$ for all $t > T^{Pre}$, and the estimator in (3) effectively targets the counterfactual series $Y_{1,t}^{0,0}$. Consequently, the estimated effects presented in Section 4.1 encompass the combined impacts of both disclosing Graves' sexual orientation and his affiliation with the LGB community during LGBT Pride Month. This can be formalized as follows:

$$\begin{aligned} \tau_t &= Y_{1,t}^{1,1} - Y_{1,t}^{0,0} \\ &= \underbrace{\left[Y_{1,t}^{1,1} - Y_{1,t}^{0,1} \right]}_{:=\tau_t^C} + \underbrace{\left[Y_{1,t}^{0,1} - Y_{1,t}^{0,0} \right]}_{:=\tau_t^L} \end{aligned} \quad (\text{D.2})$$

with τ_t^C representing the effects of the disclosure on players' preferences for Graves, and τ_t^L representing the effects of being part of the LGB community during LGBT Pride Month on players' preferences for Graves.

D.2 Separating Simultaneous Treatment Effects

The decomposition in (D.2) offers a strategy to disentangle the effects of the two treatments C_i and L_i for Graves. If we can successfully estimate the two counterfactual series $Y_{1,t}^{0,1}$ and $Y_{1,t}^{0,0}$, then we would be able to construct estimates $\hat{\tau}_t^C = Y_{1,t}^{1,1} - \hat{Y}_{1,t}^{0,1}$ and $\hat{\tau}_t^L = \hat{Y}_{1,t}^{0,1} - \hat{Y}_{1,t}^{0,0}$ of τ_t^C and τ_t^L , respectively. This would allow us to quantify the extent to which LGBT Pride Month drives the main findings of Section 4.1.

⁴⁶ Neglecting the presence of two simultaneous treatments and treating them as a single treatment does not invalidate the results of Section 4.1. It primarily affects their interpretation, which, without further investigation, could only be attributed to the combined effects of simultaneously receiving both treatments C_i and L_i - referred to as the *coming-out event* in the main body of the paper.

To this end, we assume an extended version of the SUTVA that accommodates the existence of two different treatments.

Assumption 1. (*SUTVA*): $Y_{i,t} = Y_{i,t}^{1,1} C_i L_i + Y_{i,t}^{0,1} [1 - C_i] L_i + Y_{i,t}^{0,0} [1 - C_i] [1 - L_i]$

Under Assumption 1, we can estimate the counterfactual series $Y_{1,t}^{0,0}$ by constructing a synthetic control unit that approximates the pick rates of Graves before the coming-out event as in Section 3.2. Thus, as shown in (D.2), the challenge in disentangling our causal effects of interest is to estimate $Y_{1,t}^{0,1}$ for $t > T^{pre}$, i.e., how Graves' pick rates would have evolved if Graves were already part of the LGB community prior to the 2022 LGBT Pride Month.

Having a sufficient number of LGB characters other than Graves (that is, sufficient units such as $C_i = 0$ and $L_i = 1$) would enable us to estimate the counterfactual series $Y_{1,t}^{0,1}$ through standard synthetic control methods. However, since we only have four such characters in our data set, this approach is infeasible.

One way out is to estimate the impact of LGBT Pride Month on players' preferences for LGB characters and compare the results with those obtained for Graves. If the influence of LGBT Pride Month is uniform across all LGB characters, this strategy provides insight into the role of LGBT Pride Month in driving the main findings of Section 4.1.

To achieve this, we create a composite LGB unit by averaging the pick rates of all characters such as $C_i = 0$ and $L_i = 1$ (namely, Diana, Leona, Nami, and Neeko), denoting this unit as character j without loss of generality. Then, for each period $t > T^{pre}$, we define the effect of LGBT Pride Month on players' preferences for LGB characters as the difference in character j 's potential pick rates at time t :

$$\gamma_t^L := Y_{j,t}^{0,1} - Y_{j,t}^{0,0} \quad (\text{D.3})$$

Under Assumption (1), we observe $Y_{j,t} = Y_{j,t}^{0,1}$ for all $t > T^{pre}$, and we can estimate the counterfactual series $Y_{j,t}^{0,0}$ by constructing a synthetic control unit that approximates the pick rates of character j before the beginning of the 2022 LGBT Pride Month. We can then estimate γ_t^L by computing the differences between character j 's observed pick rates and the synthetic counterfactual for all $t > T^{pre}$:

$$\hat{\gamma}_t^L = Y_{j,t}^{0,1} - \hat{Y}_{j,t}^{0,0} \quad (\text{D.4})$$

Finally, we introduce a homogeneity assumption that leverages the estimates $\hat{\gamma}_t^L$ to provide

an interpretation for the estimates $\hat{\tau}_t$ presented in Section 4.1:

Assumption 2. (*Effect Homogeneity*): $\tau_t^L = \gamma_t^L$ for all $t > T^{pre}$.

Under Assumption 2, the relationship $\tau_t^C = \tau_t - \gamma_t^L$ holds. Thus, if the estimated effects of LGBT Pride Month on players' preferences for LGB characters are small relative to the estimated effects of the coming-out event on players' preferences for Graves, this suggests that the findings of Section 4.1 must be primarily attributed to Graves' disclosure.

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